Aix-Marseille Université

Faculté des arts, lettres, langues et sciences humaines

HGABU37G01 — Geointelligence Applied to Environmental Studies

Landscape Transformation in the Yamal Peninsula, Russia

Changes in the Arctic landscape of the Yamal Peninsula through the industrial expansion of the Bovanenkovo gas field (Russia) — A temporal analysis from 1985 to 2026 and beyond

Presented by:

Anaël OLIVIER GEOFFROY • Mohamed-Ali GARCHOU • Mohamed JENNANI • Riad SALDANI • Ahmed ARHAB • A. Ezechiel SOUKOU • Abdourahmane BAKHOUM

Supervised by Pr. Sébastien GADAL

Academic Year 2025–2026

Table of Contents

Table of Contents 2
1. Introduction: Problem Statement and Study Area 4
A) General Context & Problem Statement 4
B) Presentation of the Study Area: Bovanenkovo Gas Field, Yamal Peninsula 5
Geographic Description of the Yamal Peninsula 5
Bovanenkovo Gas Field Location 6
Local Climatology 6
2. Materials and Methods 8
A) Data Acquisition 8
B) GIS Workflow 8
Preprocessing 9
Land Cover Classification 9
Index Calculation 10
C) State of the Art 10
3. Results: Territorial Evolution & Infrastructure 12
A) Historical Development (1985–2026) 12
B) Radar Analysis of Infrastructures (Sentinel-1 SAR) 17
State of the Art for SAR-Based Infrastructure Monitoring 17
Methodology 17
Results and Analysis 17
C) Permafrost Hazards & Risks 19
Objective and Physical Background of Permafrost Thaw 19
Multi-Temporal Analysis of Environmental Indices 20
Construction of the Combined RiskMAP 22
Infrastructure Exposure Map 26
4. Results: Environmental Monitoring 28
A) Thermokarst Lakes Dynamics 28
B) Vegetation Health (NDVI Analysis) 29
5. Discussion 34
A) Correlation: Gas Production vs. Territorial Expansion 34
B) Limitations & Uncertainties 35
6. Conclusion & Future Projections 37
A) General Conclusion 37
B) Future Outlook via Scenario Analysis (2026–2050) 37
Data Sources 39

References..... 40

1. Introduction: Problem Statement and Study Area

A) General Context & Problem Statement

The Arctic region plays a strategic role in the global energy system, particularly due to the large reserves of hydrocarbons located in northern Russia. Among these reserves, the Yamal Peninsula is one of the most important gas-producing regions in the world. Russia holds the largest natural gas reserves globally, and companies such as Gazprom have developed large-scale extraction infrastructures in remote Arctic territories to exploit these resources. The Bovanenkovo gas field, discovered in 1972 and developed from 1988 onwards, is today one of the largest gas production sites in the Russian Arctic.

The development of such industrial infrastructures in the Arctic raises important environmental and territorial questions. The Arctic environment is extremely fragile and sensitive to disturbances. Its ecosystems are characterized by permafrost soils, tundra vegetation, thermokarst lakes, and highly specialized biodiversity. Industrial development in these environments can lead to significant transformations of landscapes, including the construction of roads, railways, pipelines, drilling platforms, and residential facilities for workers.

Furthermore, climate change is accelerating environmental transformations in the Arctic. Rising temperatures are causing the thawing of permafrost, which destabilizes soils and modifies hydrological systems. This process can create thermokarst lakes or cause the drainage of existing water bodies. When combined with intensive industrial development, these changes can significantly alter the structure and functioning of Arctic landscapes.

Remote sensing and Geographic Information Systems (GIS) provide powerful tools to monitor these transformations over long periods of time. Satellite imagery allows researchers to observe land cover changes, vegetation health, infrastructure expansion, and hydrological dynamics even in remote and inaccessible regions such as the Yamal Peninsula. By combining different satellite datasets such as Sentinel imagery and historical observations, it becomes possible to analyze the spatial and temporal evolution of industrial territories.

This study focuses specifically on the Bovanenkovo gas field located in the central part of the Yamal Peninsula. Since the beginning of construction activities in 1988, and especially after the start of gas production in 2012, the area has experienced significant industrial expansion. This expansion has likely transformed both the natural environment and the spatial organization of the territory.

Therefore, the main research question guiding this study is the following:

> ***"How does the industrial expansion of the Bovanenkovo gas field transform the Arctic landscape of the Yamal Peninsula? A temporal analysis from 1985 to 2026."***

Through the use of satellite data and GIS analysis, this report aims to identify and quantify the territorial and environmental transformations associated with the development of this major Arctic gas field.

Key Dates:

- 1972 — Discovery of the Bovanenkovo gas field by Gazprom
- 1988 — Beginning of construction works at Bovanenkovo
- 2008 — Launch of the major industrial development phase
- 2012 — Start of gas extraction and production by Gazprom

Study Area Coordinates:

- Central point: X: 470,462 — Y: 7,818,843 (EPSG:32642, UTM Zone 42N)
- Bounding box: (X min: 470,462; Y min: 7,811,326) — (X max: 484,351; Y max: 7,829,231)

B) Presentation of the Study Area: Bovanenkovo Gas Field, Yamal Peninsula

Geographic Description of the Yamal Peninsula

The Yamal Peninsula is located in the Yamalo-Nenets Autonomous Okrug of northwest Siberia, Russia, positioned at approximately 70°40‘15”N, 70°08‘12”E. This Arctic landmass extends roughly 700 km northward into the Kara Sea, bounded by the Kara Sea and Baydaratskaya Bay to the west, and the Gulf of Ob to the east. At its northern terminus, the Malygina Strait separates it from Bely Island, while the Gyda Peninsula lies across the Ob estuary to the east. The peninsula’s name derives from the Nenets language, meaning “End of the Land,” reflecting its position at the edge of the Eurasian continent.

The peninsula’s landscape consists predominantly of continuous permafrost with numerous thermokarst lakes, the largest being Lakes Neito and Yambuto in the central region. The terrain is characterized by flat tundra plains with minimal elevation variation, creating a monotonous topography interrupted only by river valleys and lake depressions. The permafrost extends to depths of 200–300 metres in most areas, with an active layer thickness varying from 0.5 to 1.5 metres depending on local conditions and seasonal variations.

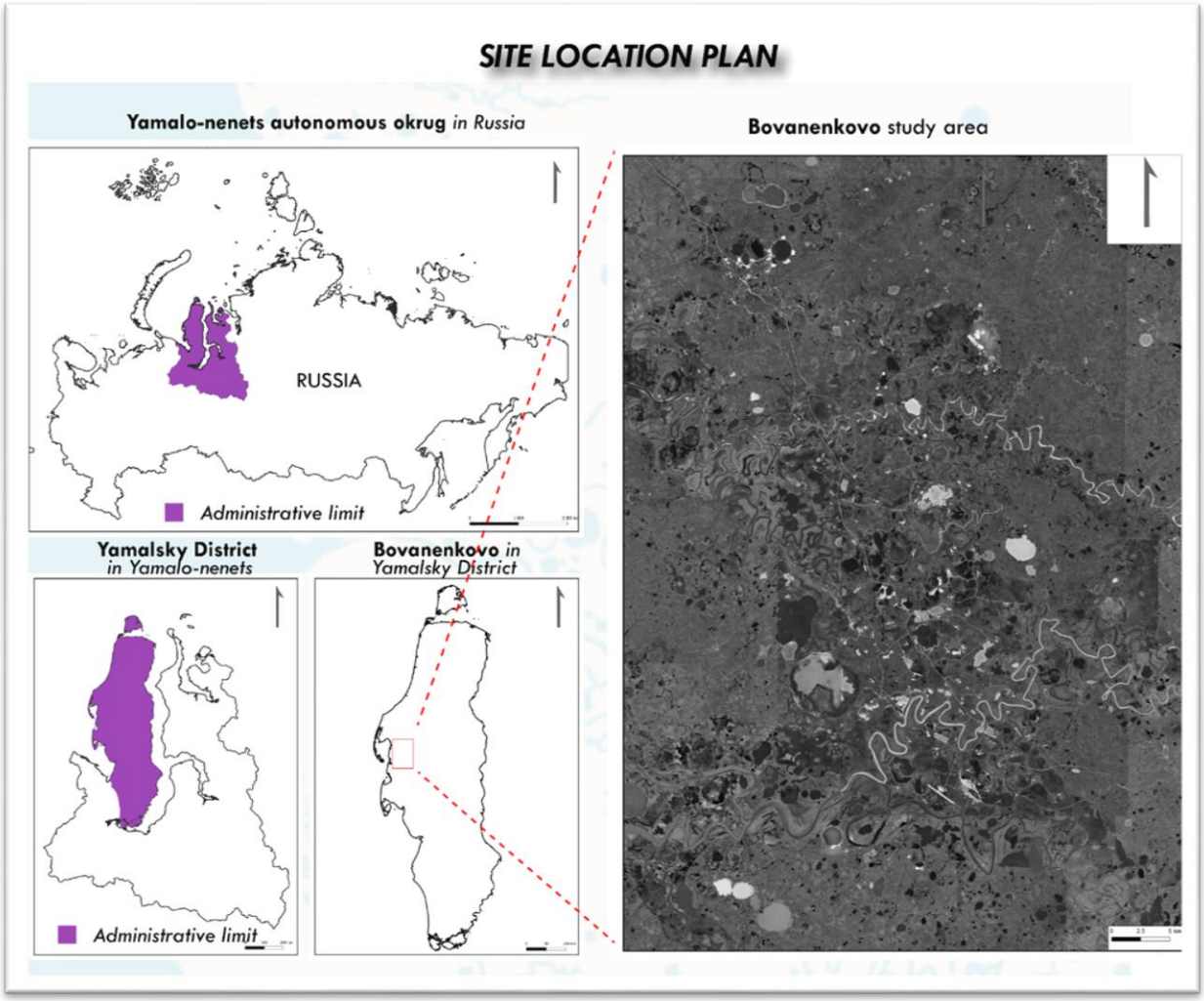


Figure 1-1. Geographic location of the Yamal Peninsula (Russia) and the Bovanenkovo gas field study area. Coordinate system: UTM Zone 42N (EPSG:32642). Scale bar: 50 km.

Bovanenkovo Gas Field Location

The Bovanenkovo gas field is situated in the central-western portion of the Yamal Peninsula at approximately 70°22'N, 68°26'E, roughly 40 km from the western coastline along the Kara Sea. This supergiant gas field represents one of the largest natural gas reserves globally, with estimated reserves exceeding 4.9 trillion cubic metres. The field extends over an area of approximately 1,200 km² and lies within the Arctic tundra zone, characterized by the complete absence of tree vegetation and dominated by low-growing shrubs, grasses, and mosses.

The site's infrastructure includes the Obskaya–Bovanenkovo railway, constructed specifically to service the gas field operations, connecting it to the mainland Russian railway network. The development area encompasses multiple drilling pads, gas processing facilities, and worker settlements, all engineered to withstand extreme Arctic conditions.

Local Climatology

The Yamal Peninsula experiences a severe Arctic climate characterized by long, extremely cold winters and short, cool summers. Winter temperatures regularly drop below −40°C, with recorded extremes reaching −60°C, while summer temperatures

rarely exceed 10°C, averaging between 3–7°C during the warmest months of July and August. The mean annual air temperature ranges from −8°C to −10°C across the peninsula.

Precipitation is relatively low, typically 200–400 mm annually, with most falling as snow during the eight-month winter period from October to May. The snow cover persists for approximately 240 days per year, reaching depths of 40–80 cm. Strong winds are characteristic throughout the year, with average speeds of 6–7 m/s, frequently exceeding 20 m/s during winter storms.

The region experiences polar night conditions from late November to mid-January, with continuous darkness for approximately 50 days at Bovanenkovo's latitude. Conversely, the polar day period extends from late May to mid-July. This extreme photoperiod variation significantly influences biological processes, permafrost dynamics, and surface energy balance, creating unique challenges for both natural ecosystems and industrial operations.

2. Materials and Methods

A) Data Acquisition

The data acquisition strategy relies on a multi-source portfolio drawn from the Copernicus programme (ESA), the USGS Landsat archive, the ArcticDEM project, OpenStreetMap, and official Gazprom statistical reports. Each product was filtered according to specific quality criteria to guarantee the robustness of the analyses — including cloud masking, radiometric quality assessment, reprojection, and spatial clipping to the study area.

Satellite / Product	Resolution	Extracted Information	Role in the Project	QA Filters
Sentinel-1 SAR (IW/EW, VV/VH)	10–40 m	σ° (backscatter), coherence	Infrastructure detection, surface disturbances, ground stability	Mode IW, VV polarisation, summer season
Sentinel-2 MSI L2A	10 m	B4, B8, B3, B11 (NDVI, MNDWI)	Vegetation, water, ecological stress (fine mapping)	SCL cloud mask
Sentinel-3 SLSTR L2 LST	~1 km	Land Surface Temperature (LST)	Heat signature, permafrost, wetlands	quality_level = 0
Sentinel-5P TROPOMI	7×3.5–7×7 km	Tropospheric NO_2, XCH_4	Air pollution (combustion, leaks)	qa_value > 0.75 (NO_2); >0.5 (CH_4)
Landsat 5 / 7 / 8	30 m	Multispectral bands (B4, B3, B2 etc.)	Historical land cover mapping (1985–2014)	Cloud cover filter, atmospheric correction
ArcticDEM	2 m	Digital Elevation Model	Topography, TWI, hillshade	Spatial filter, chronological sort
OpenStreetMap (QuickOSM)	Vector	Roads, pipelines, railways, buildings, airport	Infrastructure overlay and exposure analysis	Overpass API extraction
Gazprom (annual reports)	—	Annual gas production (bcm)	Industrial context and economic correlations	—

Table 2-1. Summary of satellite products and data sources used in this study.

B) GIS Workflow

The GIS workflow implemented in this study follows a structured multi-source approach combining optical, radar, thermal, and topographic data. The general workflow comprises three main phases: data preprocessing, feature extraction through index calculation, and classification and analysis.

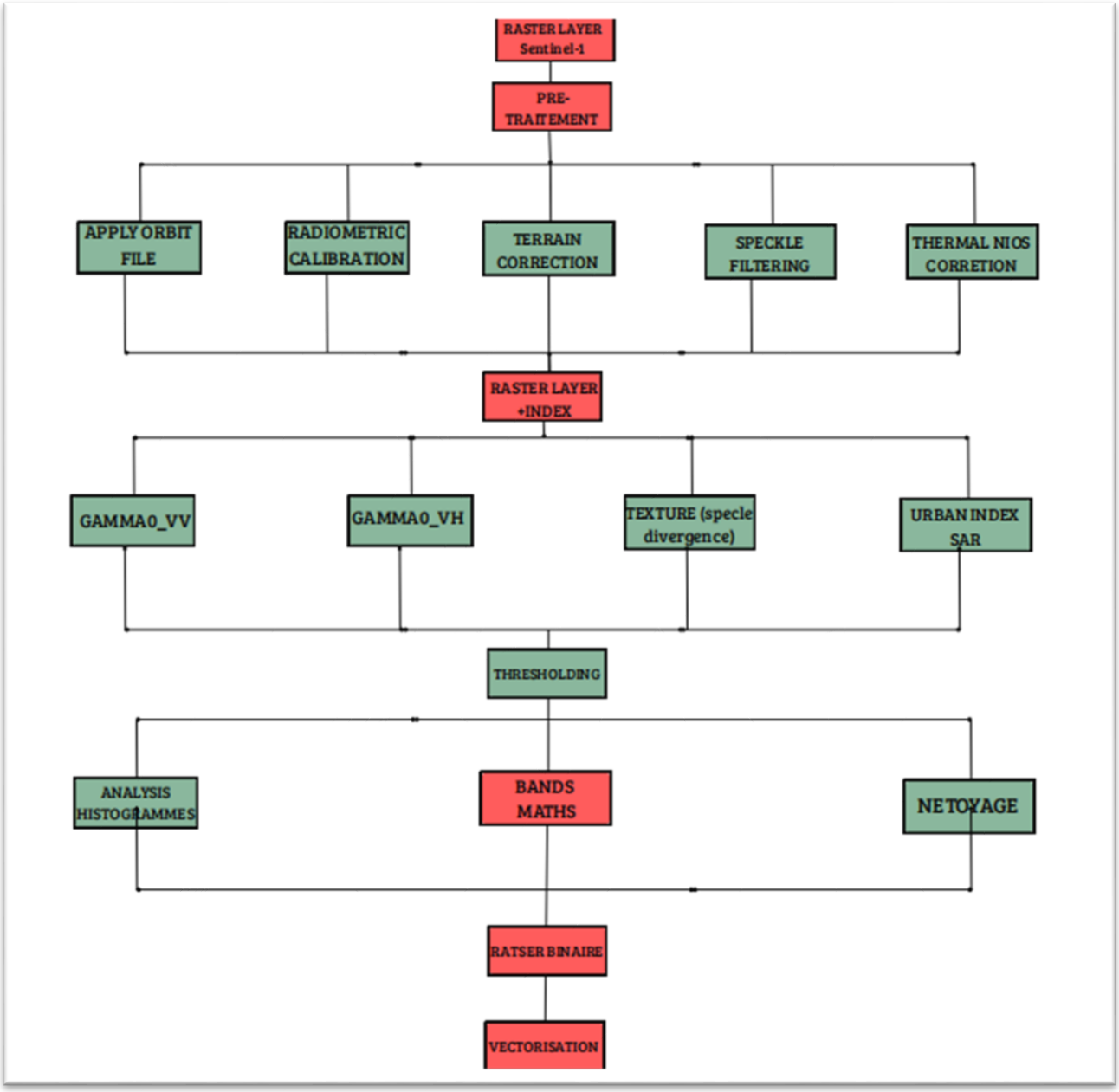


Figure 2-1. General methodological schema of the multi-source GIS workflow applied to the Bovanenkovo study area.

Preprocessing

For Sentinel-2 optical imagery (L2A), preprocessing included Scene Classification Layer (SCL) cloud masking, reprojection to EPSG:32642 (UTM Zone 42N), and spatial clipping to the study area. For historical Landsat imagery (5, 7, and 8), atmospheric correction, mosaicking, and spatial clipping were applied. Landsat 7 data between 2003 and 2013 presented specific challenges due to the failure of the Scan Line Corrector (SLC) in May 2003, which introduced systematic diagonal data gaps across all scenes from that period.

Sentinel-1 SAR data were processed through orbit correction, radiometric calibration, terrain correction (using Range Doppler Terrain Correction), speckle filtering, and thermal noise removal, following standard Sentinel Application Platform (SNAP)

workflows. A seasonal summer mean was subsequently calculated to reduce speckle noise and obtain a representative backscatter signal. ArcticDEM strips were spatially filtered, chronologically sorted, and mosaicked to produce a seamless elevation model for the 2015–2023 period, complemented by a hillshade product to improve the visual interpretation of thermokarst microforms and topographic deformations.

Land Cover Classification

Land cover classification of optical imagery was performed using the ISOCLUST unsupervised classification algorithm. A three-class scheme was adopted: (1) built-up surfaces and bare soil, (2) vegetation, and (3) water bodies. The Modified Normalized Difference Water Index (MNDWI) was preferred over the standard NDWI, as MNDWI improves the extraction of water surfaces by further reducing spectral confusion with other surface types — a particularly important distinction in the Arctic tundra context of Bovanenkovo where wet soils and shallow water bodies are spectrally similar. The number of ISOCLUST iterations was determined through systematic testing; fifteen or more iterations consistently produced reliable and stable results.

For Sentinel-1 SAR data, infrastructure detection relied on thresholding applied to Gamma0 backscatter coefficients (VV and VH polarizations), texture metrics, and a composite Urban SAR Index. Histogram-based thresholding and band mathematics were used to isolate high-backscatter structures corresponding to industrial installations such as drilling pads, processing facilities, and transport infrastructure.

Index Calculation

Multiple spectral and topographic indices were calculated across five key dates (1985, 2000, 2011, 2017, and 2023) to enable multi-temporal environmental monitoring:

- NDVI (Normalized Difference Vegetation Index): (B8 − B4) / (B8 + B4) — vegetation health and density
- MNDWI (Modified Normalized Difference Water Index): (Green − SWIR) / (Green + SWIR) — water surface detection; threshold: MNDWI > 0.20
- NDMI (Normalized Difference Moisture Index): soil moisture and active layer dynamics
- TWI (Topographic Wetness Index): computed from ArcticDEM — identification of saturated zones and thermokarst-prone sectors
- LST Anomaly (LSTA): MODIS-derived surface temperature compared to the multi-year historical average to detect thermal stress

C) State of the Art

The Bovanenkovo gas field on the Yamal Peninsula has been the subject of extensive multidisciplinary research due to the complex intersection of massive hydrocarbon extraction, a highly sensitive cryospheric environment, and traditional indigenous land use.

Understanding the subsurface dynamics of the Yamal Peninsula is critical for assessing both environmental changes and industrial risks. Research on the geosciences of the Bovanenkovo field highlights the extreme complexity of its

permafrost conditions, which include massive ground ice layers and shallow gas hydrates. The degradation of this permafrost — driven by both global climate change and localized thermal disruptions from industrial infrastructure — triggers severe geological hazards, including widespread thermokarst subsidence and explosive gas emissions that pose direct threats to pipelines and extraction facilities while fundamentally altering the regional hydrology.

The monitoring of Yamal's fragile tundra has heavily relied on Earth observation technologies. Early ecological evaluations utilizing historical remote sensing data demonstrated that different tundra ecotypes exhibit varying degrees of vulnerability to anthropogenic disturbances, establishing baselines for mapping ecological sensitivity before the most intensive phases of gas extraction began. More recently, long-term satellite tracking over decades has revealed significant land cover transformations across the Bovanenkovo region. Multi-spectral data, particularly vegetation indices like NDVI, have highlighted not only the direct physical footprint of roads and drilling pads but also the indirect degradation of surrounding vegetation caused by dust deposition, off-road vehicle tracks, and altered drainage patterns.

The industrial expansion on the Yamal Peninsula does not occur in an empty wilderness, but within an actively managed socio-ecological system. Modern spatial analyses emphasize the importance of integrating remote sensing data with local knowledge to fully capture the impacts of hydrocarbon exploitation. While high-resolution satellite imagery effectively detects large-scale infrastructural changes, ground-level observations from Nenets reindeer herders are essential for identifying diffuse pollution and subtle ecological shifts. Furthermore, this integrated approach documents the remarkable resilience and adaptation of traditional pastoralist networks navigating an increasingly fragmented industrial landscape.

The existing literature clearly establishes that single-variable monitoring is insufficient for a region as complex as Bovanenkovo. Building upon historical ecological baselines, structural cryospheric hazard assessments, and long-term spectral vegetation monitoring, this present study adopts a multi-source GIS workflow to comprehensively assess the territorial transformation of Bovanenkovo up to the year 2026.

3. Results: Territorial Evolution & Infrastructure

A) Historical Development (1985–2026)

To analyse the historical development of infrastructure related to gas exploitation at the Bovanenkovo site, land cover maps were produced with a minimum interval of five years between 1985 and 2023. Only the most relevant years were selected: 1990, 2000, 2011, 2014, and 2018. It is important to highlight the difficulty of finding usable images between 2000 and 2010. This difficulty can first be explained by the local climatic context, as the area is covered by snow for a large part of the year. It is also due to a technical failure that affected the quality of Landsat 7 images: in May 2003, the satellite's Scan Line Corrector failed, significantly degrading image quality by creating large diagonal bands of missing data.

Land cover map — 1990:

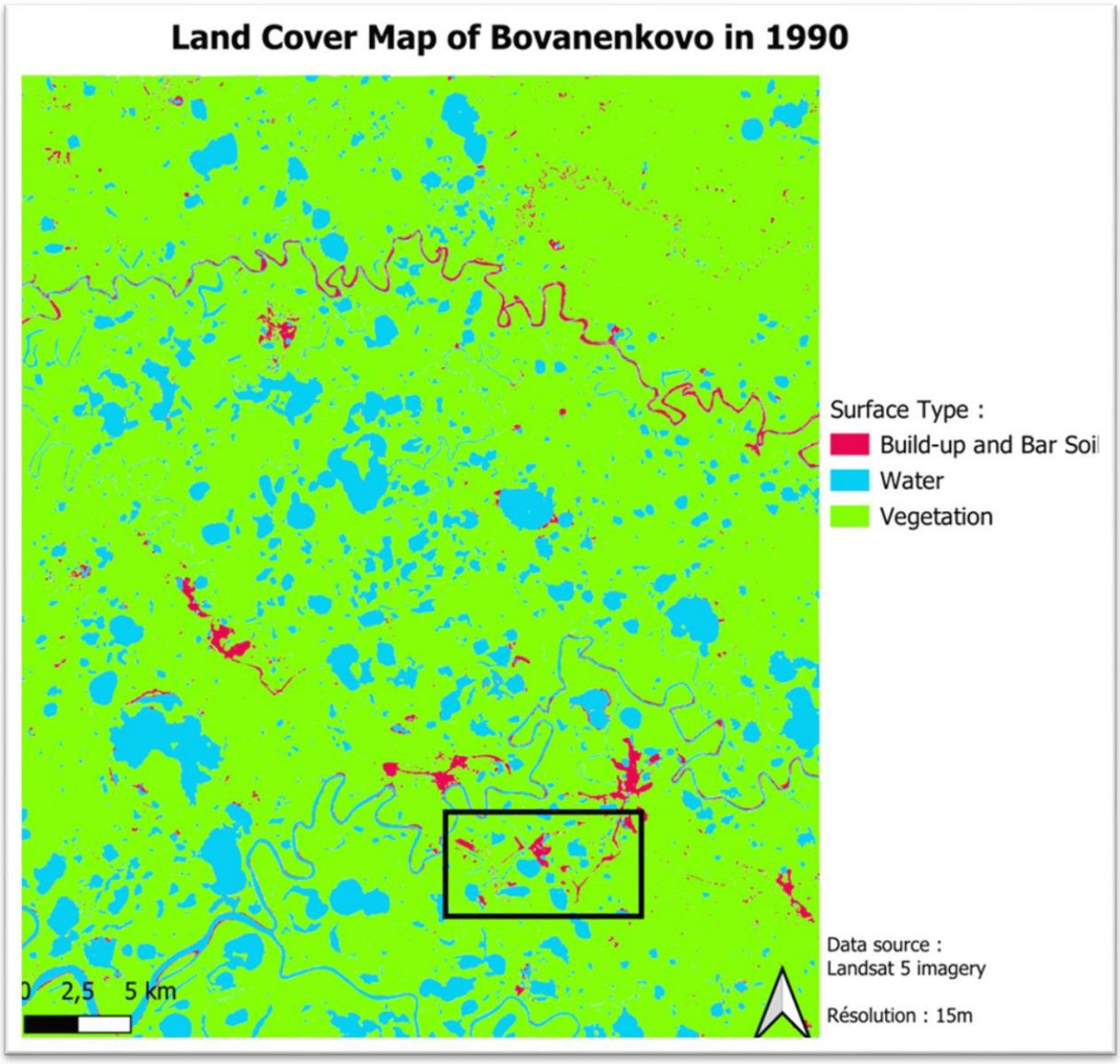


Figure 3-1. Land cover classification of the Bovanenkovo area in 1990 (Landsat, ISOCLUST). Classes: built-up/bare soil (grey), vegetation (green), water (blue).

The 1990 map shows the area to be mainly covered by vegetation and small lakes. Built-up surfaces are almost non-existent at this date. Only a few infrastructures, such as small roads, appear to have started to be developed. They are visible in the southern part of the map, within the area delimited by the reference rectangle, where they appear as relatively linear and continuous features. A few shapes that may correspond to small buildings can also be identified.

Land cover map — 2000:

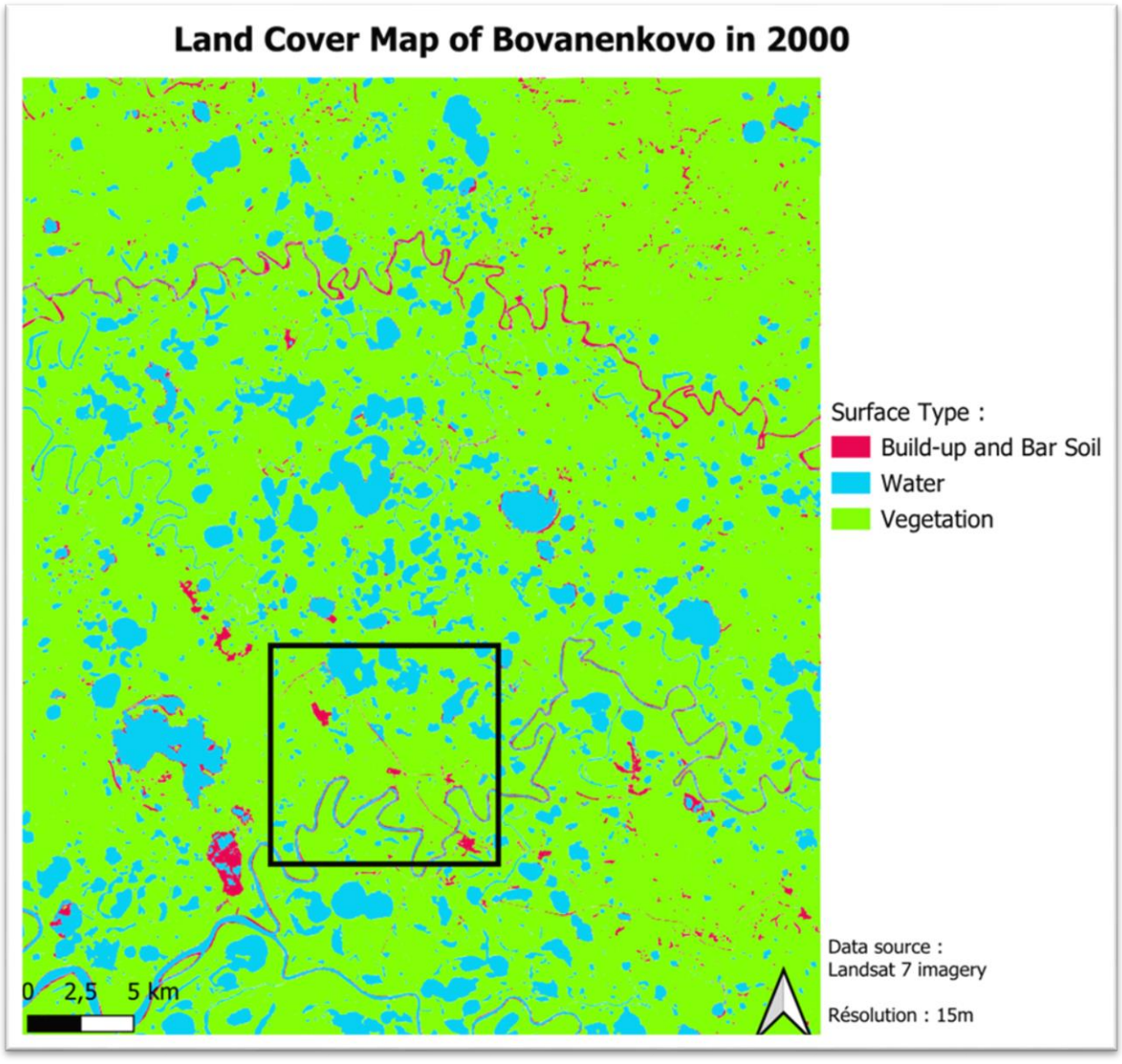


Figure 3-2. Land cover classification of the Bovanenkovo area in 2000 (Landsat).

The 2000 map shows an increase in infrastructure within the reference rectangle. These features clearly correspond to roads, identifiable by their linear and continuous shapes, as well as to built-up areas appearing as block-like forms. However, the area remains only slightly artificialised, as the major industrial development of the site only really began from 2008 onwards.

Land cover map — 2011:

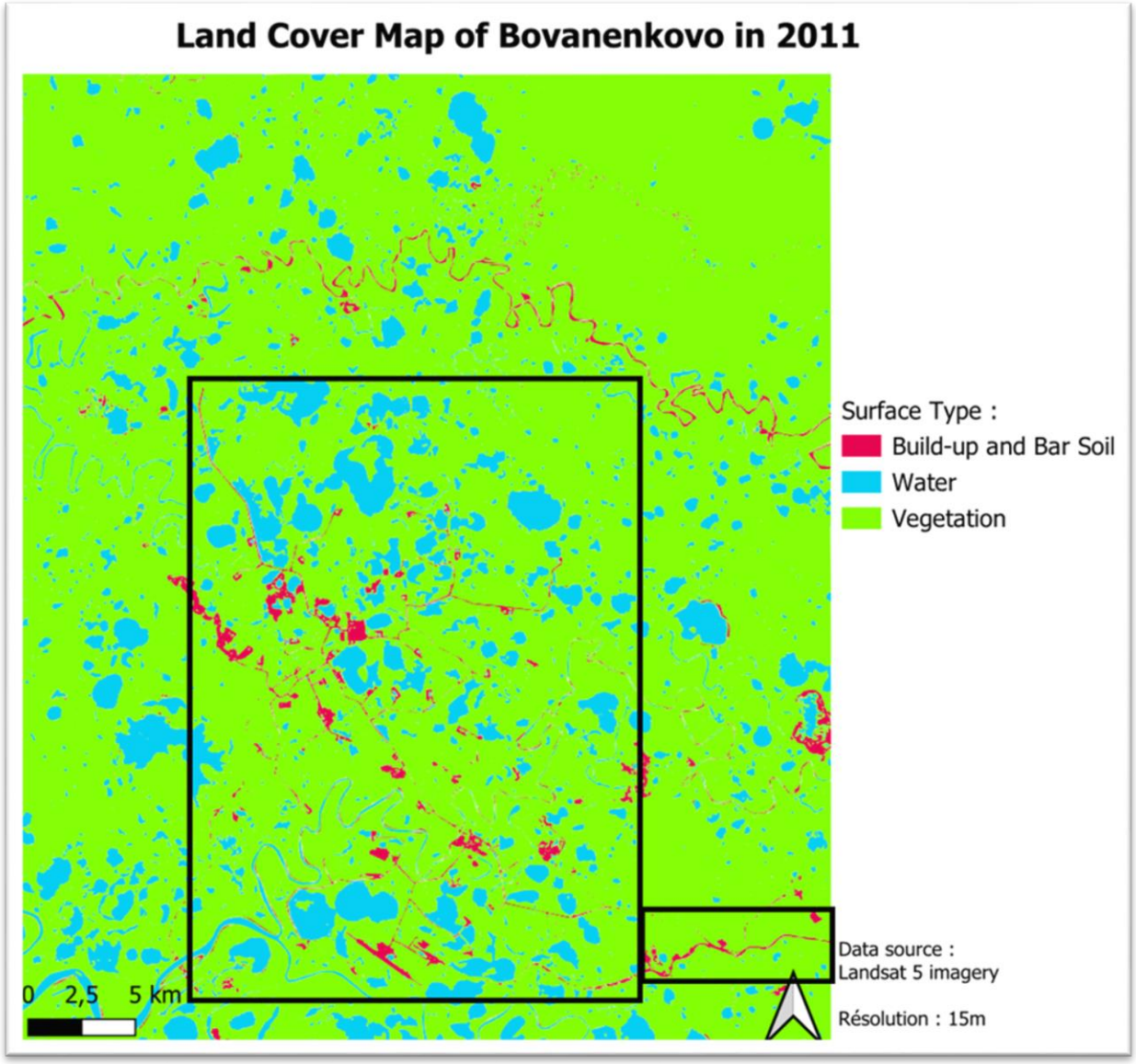


Figure 3-3. Land cover classification of the Bovanenkovo area in 2011 (Landsat). The reference rectangle highlights the main construction phase initiated in 2008.

The 2011 map highlights the appearance of numerous installations, linked to the major construction works that began in 2008. Within the reference rectangle in the centre of the map, linear and continuous built-up features can be observed, mainly corresponding to roads. Large built-up areas with relatively symmetrical shapes can also be identified, which may correspond to administrative buildings, residential facilities, or storage areas. Some built-up features — square in shape, regularly spaced along the roads — may be interpreted as gas extraction pads.

South of this rectangle, a large built-up area stands out due to its characteristic shape, suggesting it corresponds to an airport. Near this airport, a logistics station can also be identified, recognizable through the presence of a continuous built-up surface extending from the station towards the eastern edge of the map, which probably corresponds to the railway line.

Land cover map — 2014:

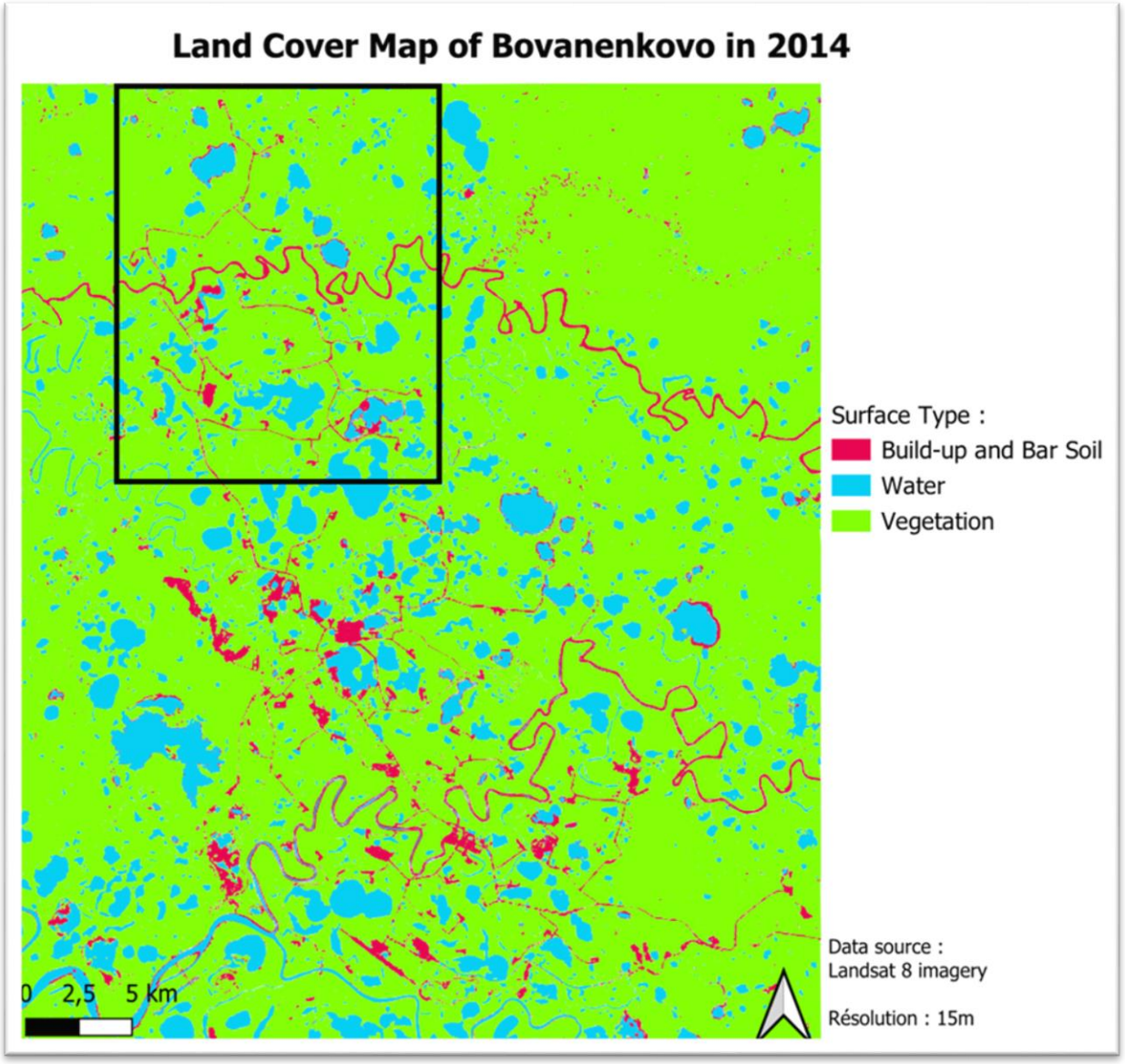


Figure 3-4. Land cover classification of the Bovanenkovo area in 2014 (Landsat).

The 2014 map shows that significant infrastructure expansion occurred after 2011, mainly towards the north. This expansion concerns primarily roads — identifiable by their linear and relatively continuous shapes — as well as extraction sites recognisable by the small size of their built-up surfaces. The main logistics sites, however, remain mostly concentrated in the southern part of the map. When considering the map as a whole, a probable overrepresentation of bare soils compared to actual built-up surfaces can be observed, linked to classification limitations.

Land cover map — 2018:

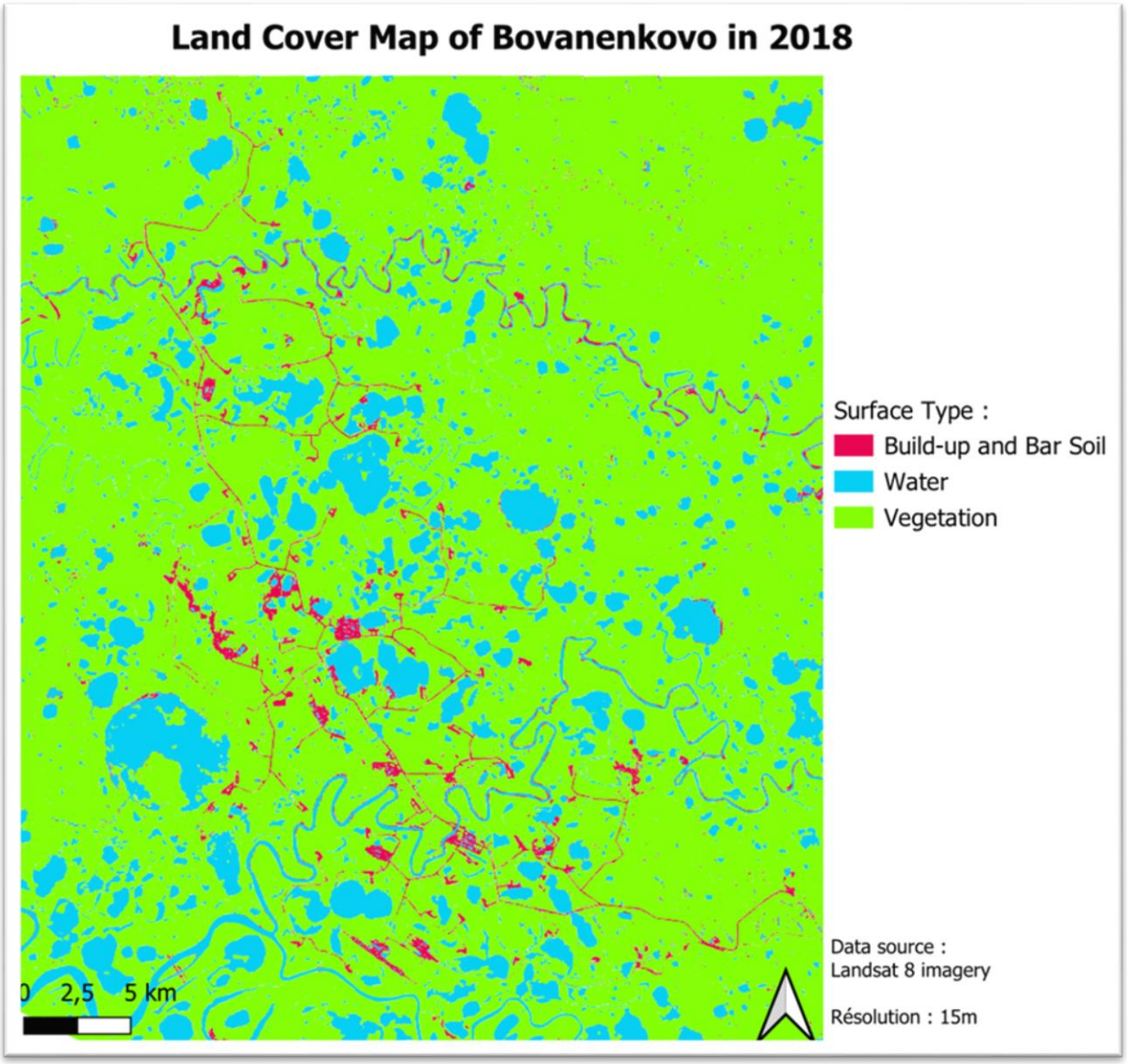


Figure 3-5. Land cover classification of the Bovanenkovo area in 2018 (Landsat).

The 2018 map makes it possible to identify built-up areas more clearly than on the previous map, as bare soils are less represented. The built-up areas appear very similar to those already observed in 2014, suggesting that infrastructure expansion between 2014 and 2018 was very limited or almost non-existent.

To summarise, the analysis of the land cover maps demonstrates that in 1990, built-up areas were almost non-existent at Bovanenkovo. By 2000, a few infrastructures had started to appear, mainly in the form of roads and small built-up areas. In 2011, most of the built-up surfaces became visible, following the major phase of industrial development carried out by Gazprom from 2008 onwards. These areas include the main facilities related to housing, storage, and logistics, with the presence of a railway station and an airport. In 2014, infrastructure expansion continued northward, mainly through the development of extraction sites. Finally, in 2018, industrial development appears to have been very limited compared with 2014, suggesting a relative

stabilisation of the infrastructure footprint.

B) Radar Analysis of Infrastructures (Sentinel-1 SAR)

State of the Art for SAR-Based Infrastructure Monitoring

The use of Sentinel-1 has become a reference approach in remote sensing for monitoring land surface dynamics, particularly in challenging environments such as the Arctic. Based on Synthetic Aperture Radar (SAR), Sentinel-1 provides all-weather, day-and-night observations, making it especially suitable for regions affected by frequent cloud cover and long periods of darkness. Scientific studies have shown its strong capability to detect anthropogenic features, notably built-up areas and linear infrastructure, due to their high radar backscatter compared to natural surfaces. Standard methodologies generally include radiometric calibration, speckle filtering, and terrain correction, followed by the analysis of backscatter coefficients (VV and VH polarizations) and texture measures to enhance surface discrimination. Classification techniques often rely on thresholding or supervised algorithms to extract features of interest. However, despite its robustness, Sentinel-1 presents limitations, particularly in detecting small or low-contrast objects and in dealing with signal variability caused by soil moisture, snow cover, and surface roughness.

Methodology

The methodology applied to the Bovanenkovo gas field is based on the processing of Sentinel-1 SAR data through a structured workflow combining preprocessing, feature extraction, and classification, aimed at detecting industrial infrastructure such as the Obskaya–Bovanenkovo railway and drilling pads. The preprocessing phase — including orbit correction, radiometric calibration, terrain correction, speckle filtering, and thermal noise removal — ensures that the radar signal is physically reliable and comparable over time. The extraction of multiple indicators (Gamma0 VV, Gamma0 VH, texture metrics, and an Urban SAR Index) allows for robust differentiation between natural tundra and artificial surfaces, capturing both surface roughness and the spatial organization typical of industrial installations. Thresholding based on histogram analysis and band mathematics provides an efficient method for isolating high-backscatter structures, although it remains sensitive to environmental variability such as moisture and snow cover. Post-processing steps including cleaning, binary raster generation, and vectorization enable accurate quantification of infrastructure through surface and length measurements.

Results and Analysis

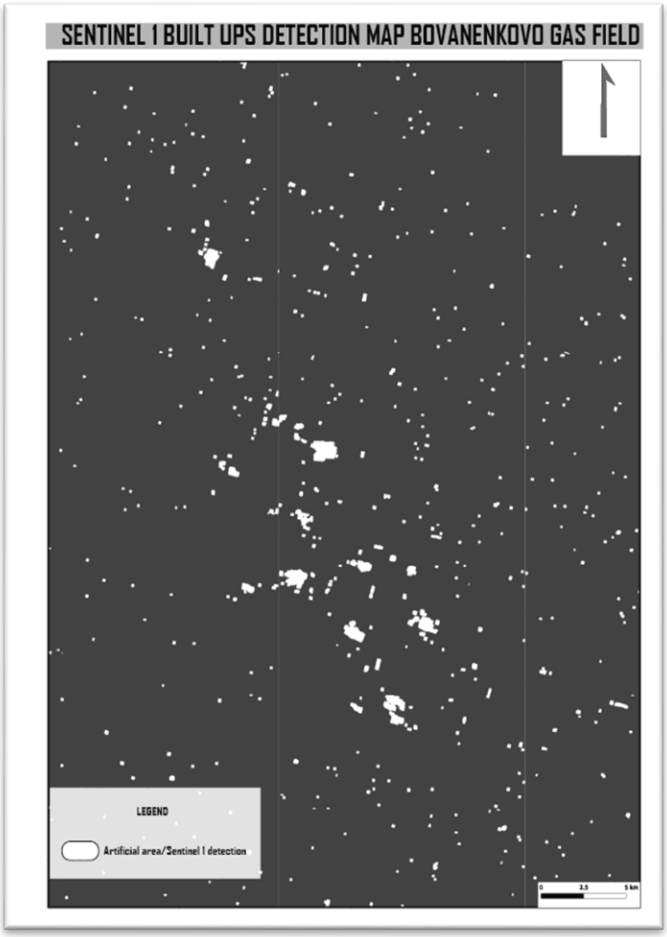


Figure 3-6. Sentinel-1 SAR backscatter classification of the Bovanenkovo gas field. High-backscatter zones (red/orange) indicate industrial built-up areas and major infrastructure. Coordinate system: UTM Zone 42N.

Global Results:

The results mainly highlight the detection of built-up structures within the Bovanenkovo gas field, such as drilling pads and major infrastructure, which appear clearly due to their strong radar backscatter. However, it was more difficult to reliably identify other components (e.g. minor roads, pipelines, or subtle surface disturbances), as these features generate weaker or more variable signals that can be confused with natural tundra variations or affected by speckle noise and seasonal conditions. This limitation highlights both the strength of SAR data for detecting large, highly reflective structures and its constraints when analysing finer or less contrasted elements of the industrial landscape.

Gas Infrastructure Spatial Organisation:

The gas infrastructure detected in the Bovanenkovo gas field shows a clearly structured and hierarchical spatial organisation strongly influenced by the pipeline network. The results highlight a central industrial core where most built-up areas are concentrated, surrounded by multiple secondary clusters of facilities distributed along branching pipeline corridors. This linear and radial pattern reflects the functional logic of gas extraction and transport, where infrastructure expands outward from main processing hubs. The detected built-up areas correspond mainly to high-backscatter

zones interpreted as gas extraction pads and processing-related installations. In contrast, more subtle components such as pipelines and small service structures are only partially detected or fragmented, due to their lower radar response and spatial resolution limitations. The map also reveals a progressive densification of industrial activity near central hubs, indicating a concentration of extraction and treatment functions in key operational nodes.

Environmental Context:

The environmental analysis of the Bovanenkovo gas field highlights a landscape strongly impacted by industrial development, where natural tundra surfaces are increasingly fragmented by extraction and transport infrastructure. The map shows a clear contrast between relatively undisturbed peripheral areas and highly modified central zones, where built-up structures, roads, and pipelines dominate. This fragmentation indicates a progressive artificialisation of the Arctic environment, reducing the continuity of natural habitats. A key environmental factor in this region is the presence of continuous permafrost, which strongly influences both surface stability and hydrological processes. The degradation of permafrost, partly driven by infrastructure construction and thermal disturbance, can lead to ground instability, surface subsidence, and the formation of thermokarst features.

Transport Network Analysis:

The transport network within the Bovanenkovo gas field reveals a highly structured system dominated by the spatial interaction between access roads, pipelines, and the railway infrastructure. Transport routes are strongly concentrated around central industrial hubs, where the density of connections is highest, indicating functional nodes of extraction and processing activity. From these core areas, a branching network of access roads extends outward, progressively linking more isolated drilling pads and operational zones. The Obskaya–Bovanenkovo railway acts as the main structural backbone of the system, organizing both spatial accessibility and industrial expansion. Peripheral areas remain weakly connected, highlighting a strong gradient of accessibility between core and marginal zones.

Overall, the integrated analysis of gas infrastructure, transport networks, and environmental conditions reveals a highly structured and interdependent industrial system in which gas exploitation, transport infrastructure, and environmental transformation are tightly interconnected, leading to a progressive and spatially organized industrialization of the Arctic landscape.

C) Permafrost Hazards & Risks

Objective and Physical Background of Permafrost Thaw

The objective of this component of the analysis is to assess permafrost degradation dynamics, thermokarst processes, hydrological modifications, and the impact on gas and oil infrastructure at Bovanenkovo, through a multi-sensor approach combining optical, radar, topographic, and infrastructure data. Two key outputs were produced: a combined permafrost sensitivity risk map (RiskMAP) and an infrastructure exposure map.

Permafrost thaw is a complex process driven by the transfer of thermal energy from the atmosphere to the subsurface, triggering a phase change in water. In Arctic environments, this phenomenon is not limited to a simple temperature increase but involves precise thermodynamic mechanisms.

Surface Energy Balance:

Thaw begins when the net energy balance at the soil surface becomes positive. The driving heat flux G — dependent on the thermal conductivity of the surface material (snow, vegetation, or bare soil) — propagates downward into the ground. The thermal conductivity of ice is approximately four times greater than that of liquid water, meaning that cold penetrates the soil more efficiently in winter than heat does in summer. As thaw progresses, the massive latent heat of fusion ($L_2 \approx 334$ kJ/kg) must be consumed to melt ground ice before the soil temperature can rise above 0°C (the "zero-curtain effect"). Once liquid water dominates, the physical properties of the terrain are radically altered.

Active Layer and Thermokarst Dynamics:

Thaw progresses from the surface downward, creating the active layer — the seasonally frozen zone above the permanently frozen ground. When ground ice (massive ice or ice lenses) melts, volume decreases by approximately 9%, causing surface subsidence, topographic deformation, and the formation of waterlogged depressions. This thermal subsidence is the foundation of the thermokarst process. The underlying permafrost acts as an impermeable floor, saturating the active layer with meltwater. This increased soil moisture modifies thermal emissivity and vegetation/moisture indices, as free water absorbs more radiation in the shortwave infrared (SWIR), providing remote sensing indicators of surface instability.

Multi-Temporal Analysis of Environmental Indices

A multi-temporal analysis was conducted across five key dates: 1985, 2000, 2011, 2017, and 2023. Spatial variations were studied through raster comparison, statistical analysis, and cartographic overlay of NDVI, NDMI, and NDWI.



Figure 3-7. Multi-temporal evolution of NDVI, NDMI, and NDWI indices at Bovanenkovo between 1985 and 2023 (Landsat imagery). The 2011–2017 period marks a critical environmental tipping point.

The evolution of environmental indices reveals three distinct phases:

Phase 1 — Relative Stability (1985–2011):

During this phase, NDMI increases progressively, reaching its maximum around 2011, suggesting an increasingly saturated tundra and vegetation rich in water — a sign of the gradual warming of permafrost releasing moisture into the active layer. NDWI decreases slightly and remains very low (around −0.20), indicating no major expansion of open water surfaces during this period. NDVI follows a broadly similar trend, reflecting a relatively stable vegetation cover.

Phase 2 — Critical Tipping Point (2011–2017):

This is the most spectacular period. A sharp rise in NDWI — from −0.19 to −0.04 — reflects the massive appearance of thermokarst lakes and hydrological expansion. Subsurface ice melts, the ground collapses, and depressions fill with water. Simultaneously, NDMI drops sharply from +0.04 to −0.23: once water accumulates in depressions, surrounding tundra zones may appear paradoxically drier by drainage. This period also coincides with the most intensive phase of industrial development at Bovanenkovo, where the replacement of wet tundra by infrastructure (roads, platforms, embankments) causes a radical drop in surface moisture indices.

Phase 3 — New Equilibrium (2017–2023):

The landscape appears to stabilize in a degraded state. NDWI remains elevated: thermokarst processes do not recede, and the lakes formed are now a permanent component of the landscape. NDMI stagnates at a very low level, confirming that the modification of land cover through industrialization and the degradation of the permafrost active layer are now irreversible at a human timescale.

Regarding Land Surface Temperature anomalies (LSTA), the normalized LST map comparing 2023 temperatures against the 2011–2023 mean reveals exclusively positive anomalies. Minimum values — located in the northern part of the study area — approach 0°C, while maximum values — concentrated in the central zone — exceed +4°C. This confirms a clear warming trend over a relatively short time window.

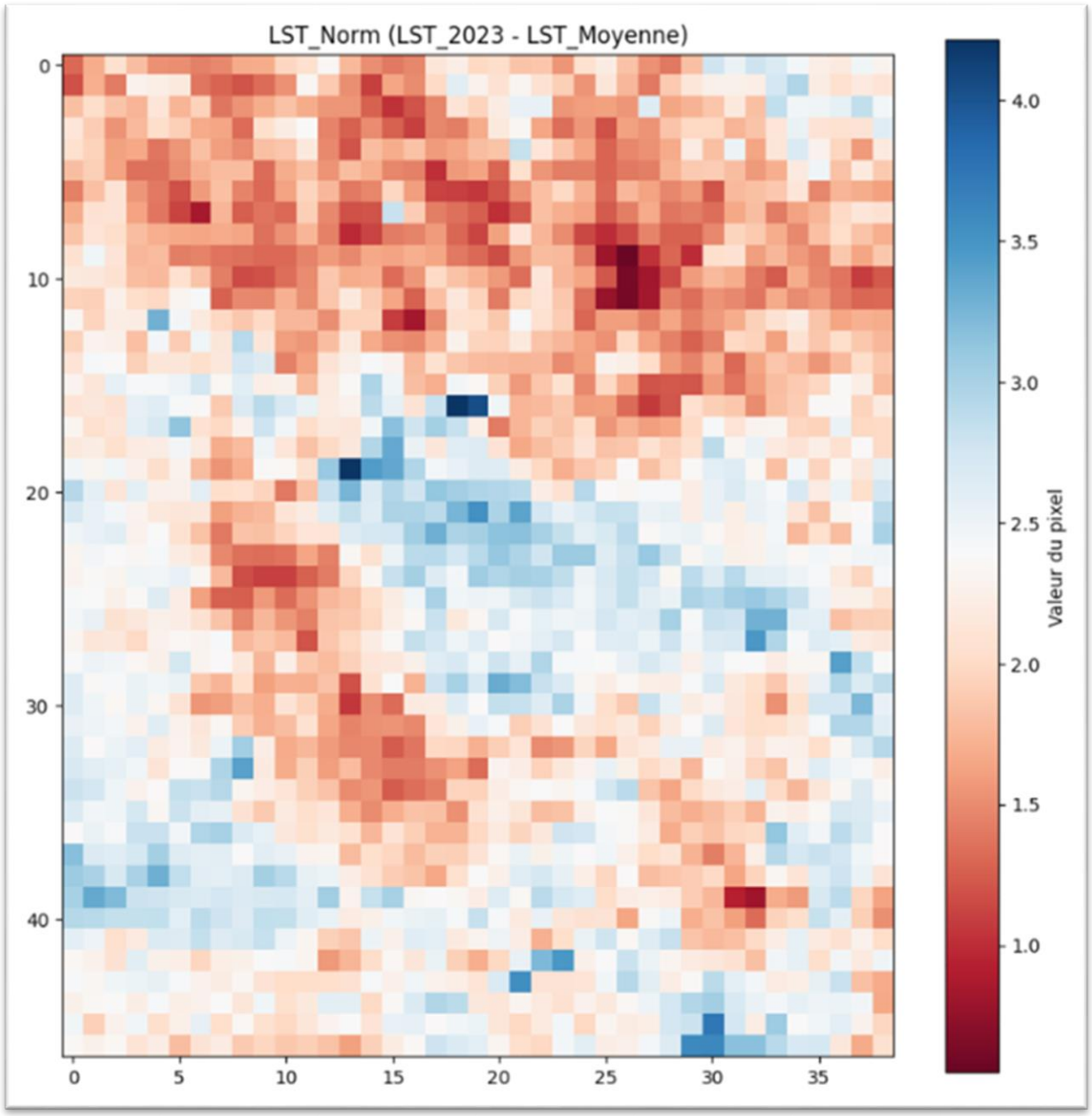


Figure 3-8. Land Surface Temperature anomaly (LSTA) map. Values represent the difference between the 2023 seasonal mean and the 2011–2023 historical mean (MODIS data, °C).

Construction of the Combined RiskMAP

A synthetic permafrost sensitivity map (RiskMAP) was constructed through multi-criteria combination of environmental indicators. The variables integrated are: TWI, NDMI, NDWI, Sentinel-1 backscatter VV, and terrain slope. All variables were normalized, reclassified, and weighted according to their influence on permafrost

stability. Three intermediate risk maps were first produced before being combined into a final composite index:

Thermokarst RiskMap (RT):

$$RT = 0.4 \times NDMI_Norm + 0.3 \times NDWI_Mask + 0.3 \times TWI_Norm$$

The RT map estimates zones most susceptible to thermokarst degradation, including soil subsidence, formation of thermokarst ponds, terrain destabilization, and infrastructure deformation. High-value areas (blue tones) correspond to humid depressions, poorly drained sectors, and zones adjacent to hydrographic networks. Orange-red areas reflect drier, better-drained sectors less susceptible to thermokarst.

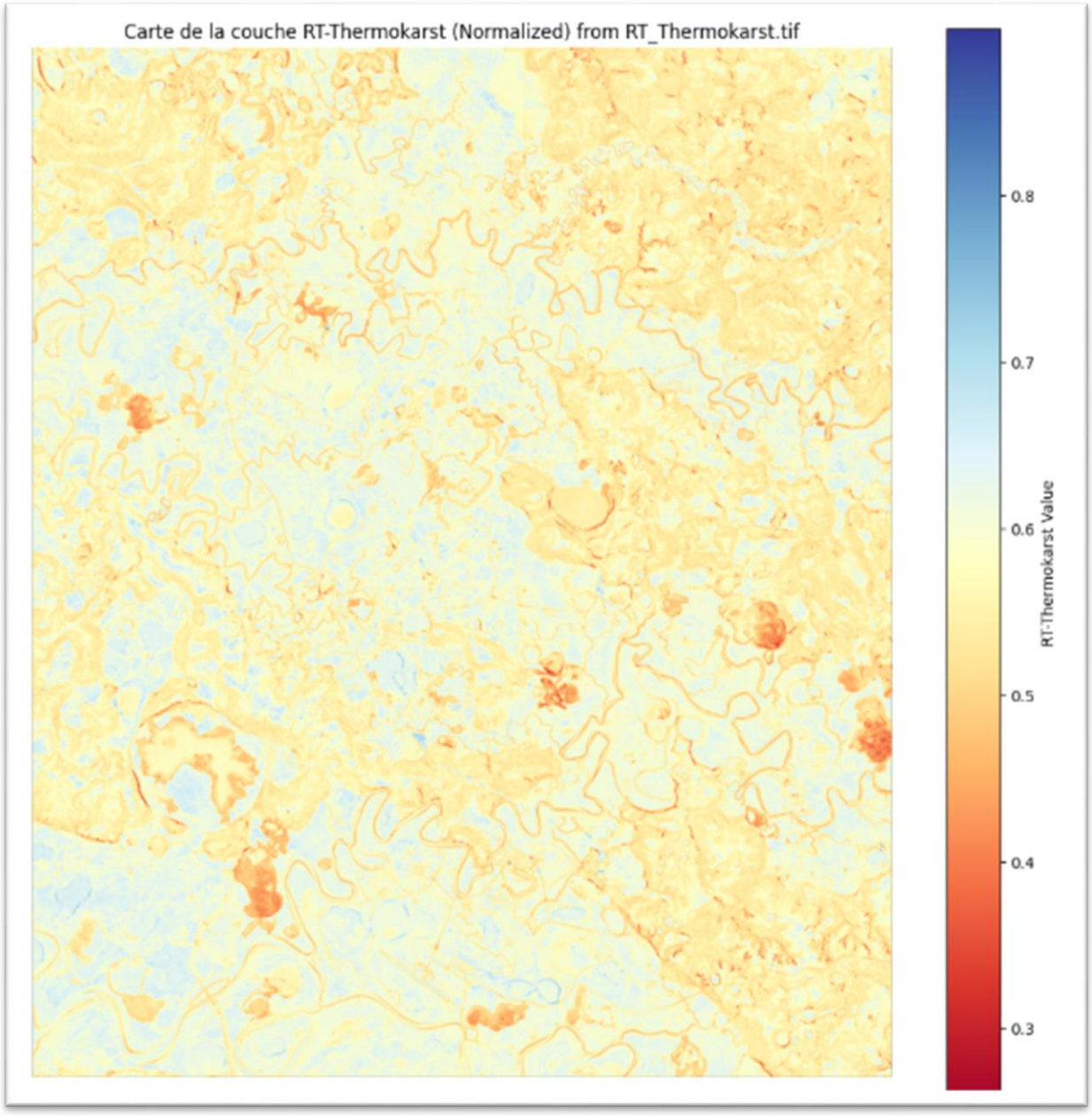


Figure 3-9. Thermokarst risk map (RT) for the Bovanenkovo study area. High values (blue) indicate susceptibility to permafrost degradation and thermokarst formation.

Subsidence RiskMap (RS):

$$RS = 0.6 \times SAR_norm + 0.4 \times LST_stress$$

The RS map estimates susceptibility to ground subsidence linked to permafrost degradation, combining SAR-derived surface deformation indicators (60%) with thermal stress from surface temperature (40%). High-value zones appear as dispersed nuclei in the central and southern parts of the study area, indicating locations where ground movement and permafrost thaw conditions are simultaneously observed. Blue-tinted zones associated with lakes and water surfaces present relatively lower values.

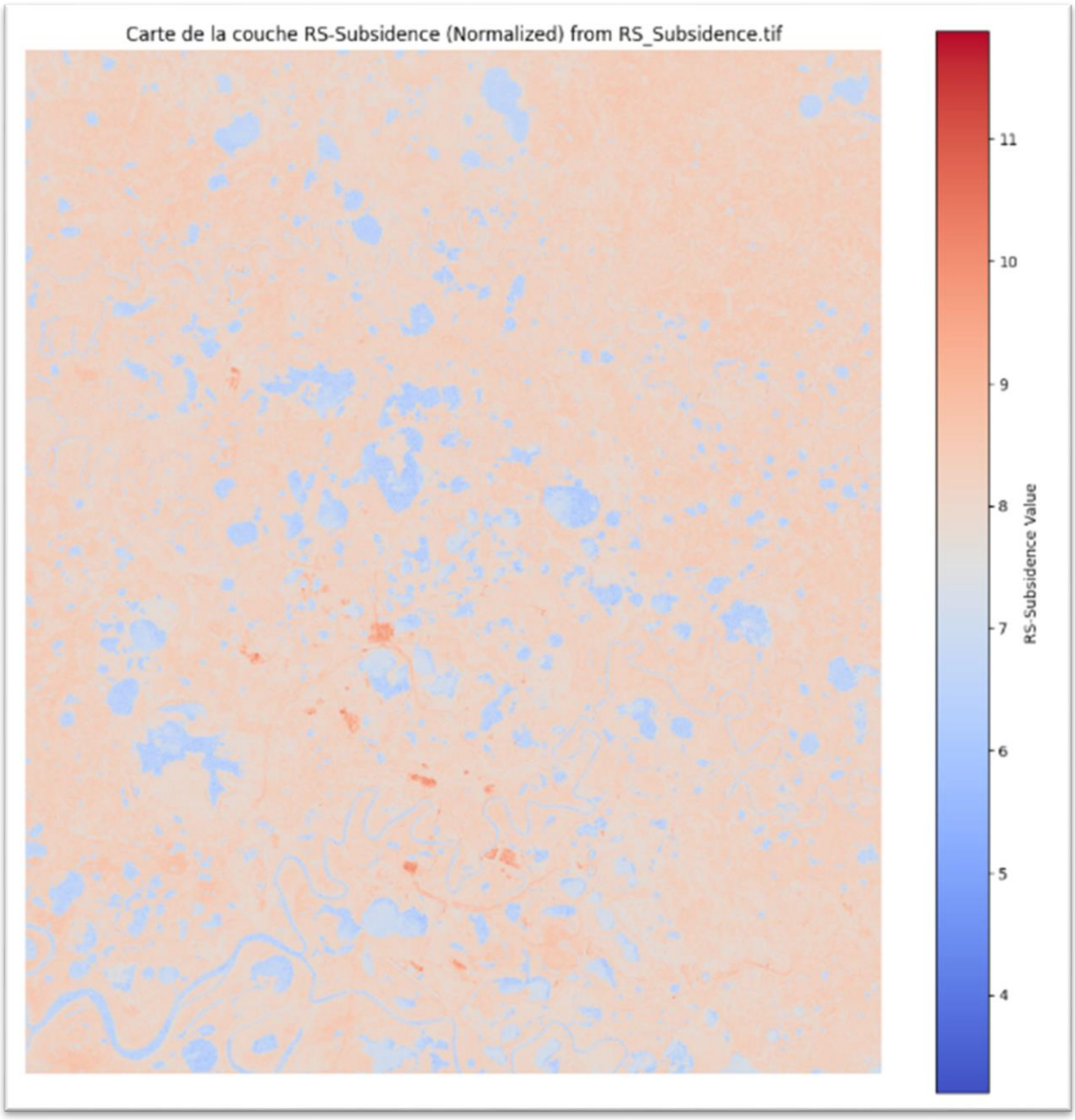


Figure 3-10. Subsidence risk map (RS) for the Bovanenkovo study area. High values indicate sectors with simultaneous ground movement and thermal conditions favorable to permafrost thaw.

Active Layer RiskMap (R_ALT):

$$R_ALT = 0.5 \times LST_norm + 0.3 \times (1 - NDVI_norm) + 0.2 \times Roughness_norm$$

The R_ALT map represents the vulnerability of the permafrost active layer to thermal warming, combining thermal stress (dominant factor at 50%), low vegetation density (1−NDVI), and terrain roughness. High-value zones — concentrated in the central, southeast, and southwest portions — correspond to sectors of warming, sparse vegetation, and irregular surface morphology, indicating increased sensitivity to permafrost degradation processes.

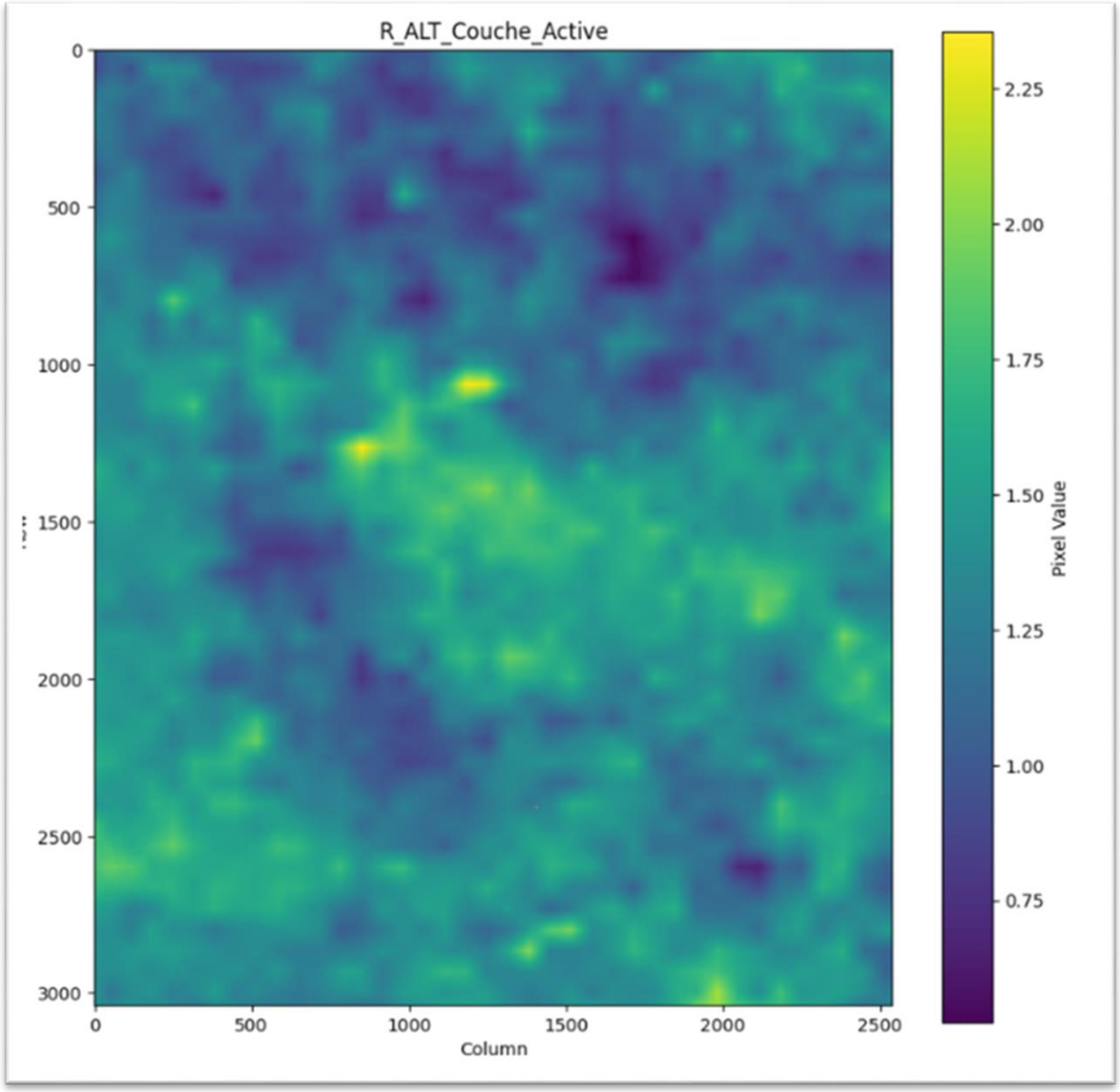


Figure 3-11. Active layer vulnerability map (R_ALT). High values (yellow-green) indicate thermally stressed, poorly vegetated, and morphologically irregular sectors most susceptible to active layer deepening.

Combined RiskMAP (RC):

$$RC = 0.35 \times RT + 0.35 \times RS + 0.35 \times R_ALT$$

The final Combined RiskMAP integrates the three intermediate risk maps with equal weighting to provide a comprehensive assessment of overall permafrost vulnerability. The spatial analysis reveals a contrasted distribution: orange and red tones indicate areas where thermokarst, subsidence, and active layer degradation risks accumulate, concentrated primarily in the central and southern portions of the study area. Green-toned areas — mainly in the north — present lower risk levels.

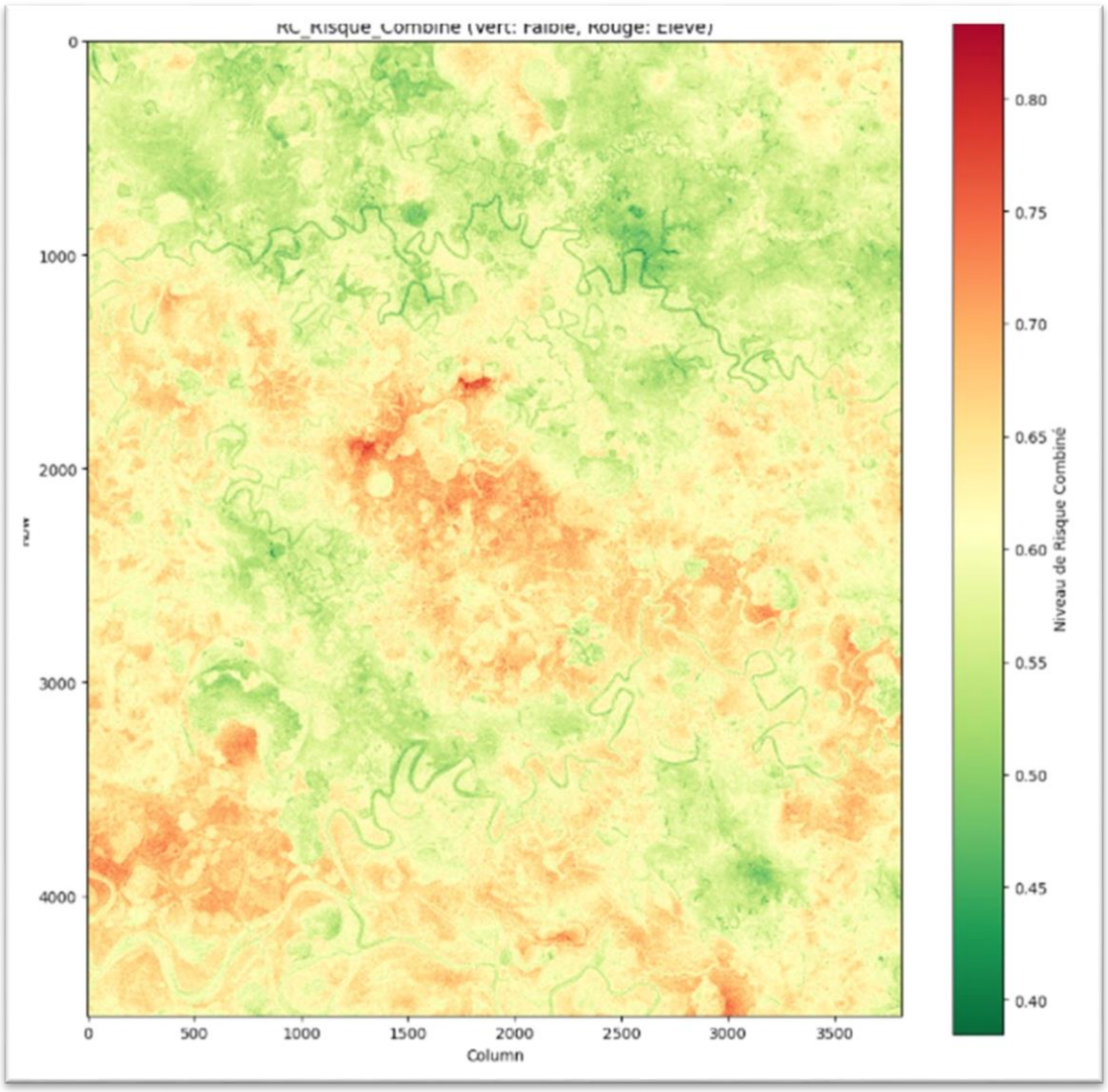


Figure 3-12. Combined permafrost risk map (RC) for the Bovanenkovo study area. Red and orange zones indicate high cumulative vulnerability; green zones indicate relatively stable permafrost conditions.

Infrastructure Exposure Map

A spatial infrastructure exposure analysis was conducted by intersecting OpenStreetMap infrastructure layers (roads, pipelines, railways, industrial zones, buildings, airports, and energy platforms) with the RiskMAP and high-risk thermokarst

and hydrological anomaly zones. The analysis includes proximity buffers, infrastructure density calculations, and intersection with the Combined RiskMAP.

The results reveal a strong concentration of risk along the infrastructure network, with the highest intensity in the central zone of the study area. The deepest red tones appear around the main infrastructure nodes and densest transport axes, indicating the spatial accumulation of exposed infrastructure. Peripheral and more isolated sections exhibit more moderate exposure levels. Green sectors indicate the absence of infrastructure and therefore zero infrastructure risk. The continuity of high-risk zones suggests that the infrastructure network is interconnected within sectors particularly sensitive to permafrost deformation.

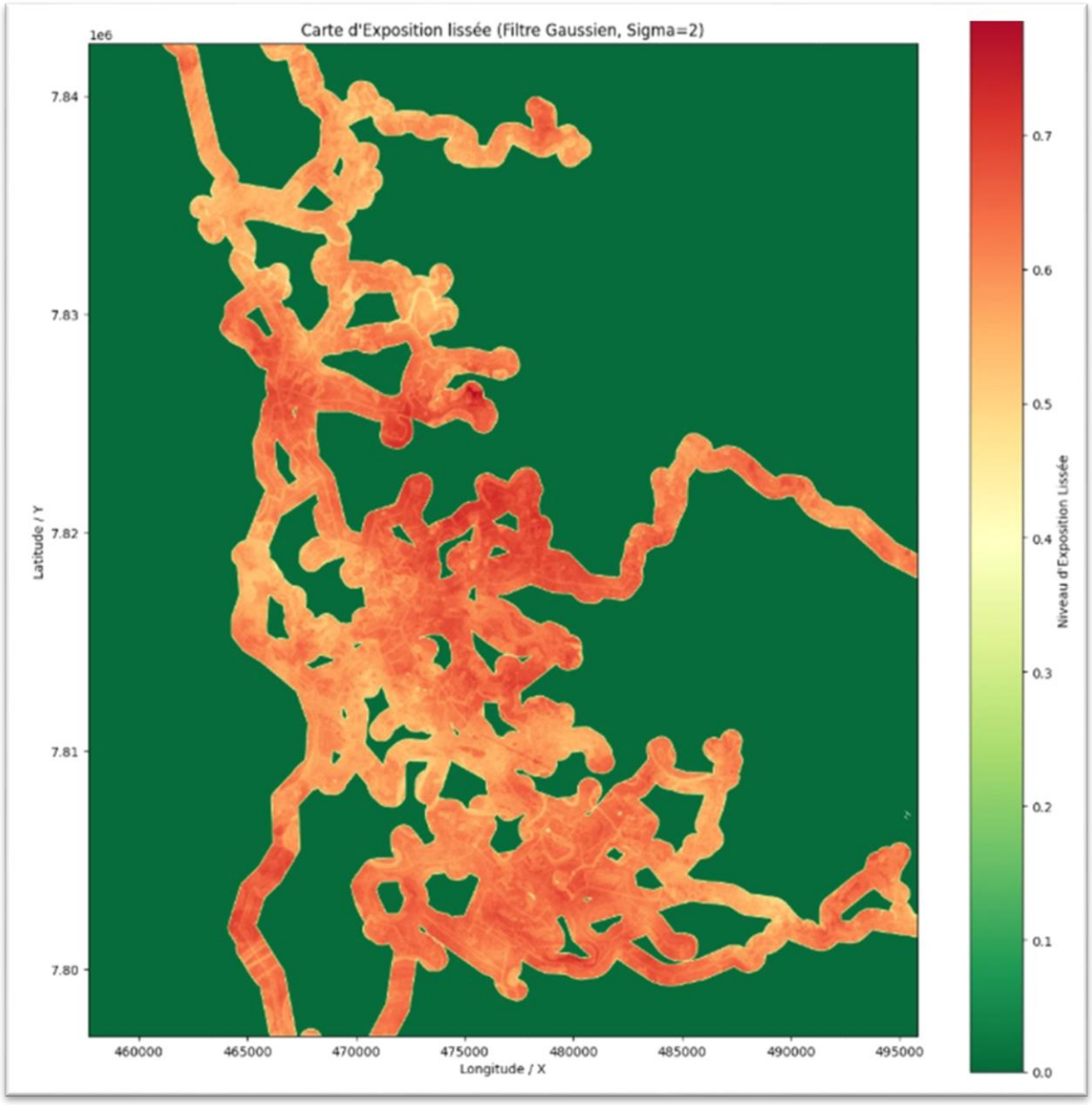


Figure 3-13. Infrastructure exposure map for the Bovanenkovo gas field. Colours indicate levels of exposure of transport and industrial infrastructure to permafrost degradation risk zones. Data sources: OpenStreetMap and Combined RiskMAP (RC).

4. Results: Environmental Monitoring

A) Thermokarst Lakes Dynamics

Methodology:

- Data: Sentinel-2 (Green + SWIR bands), summer scenes (June–August)
- Preprocessing: L2A atmospheric correction, SCL cloud mask, reprojection to EPSG:32642, spatial clip
- MNDWI calculation: (Green − SWIR) / (Green + SWIR); water detection threshold: MNDWI > 0.20
- Vectorization of water polygons, surface area calculation by date, topological control
- Multi-year surface comparison (Δ surfaces): expansion vs. drainage dynamics

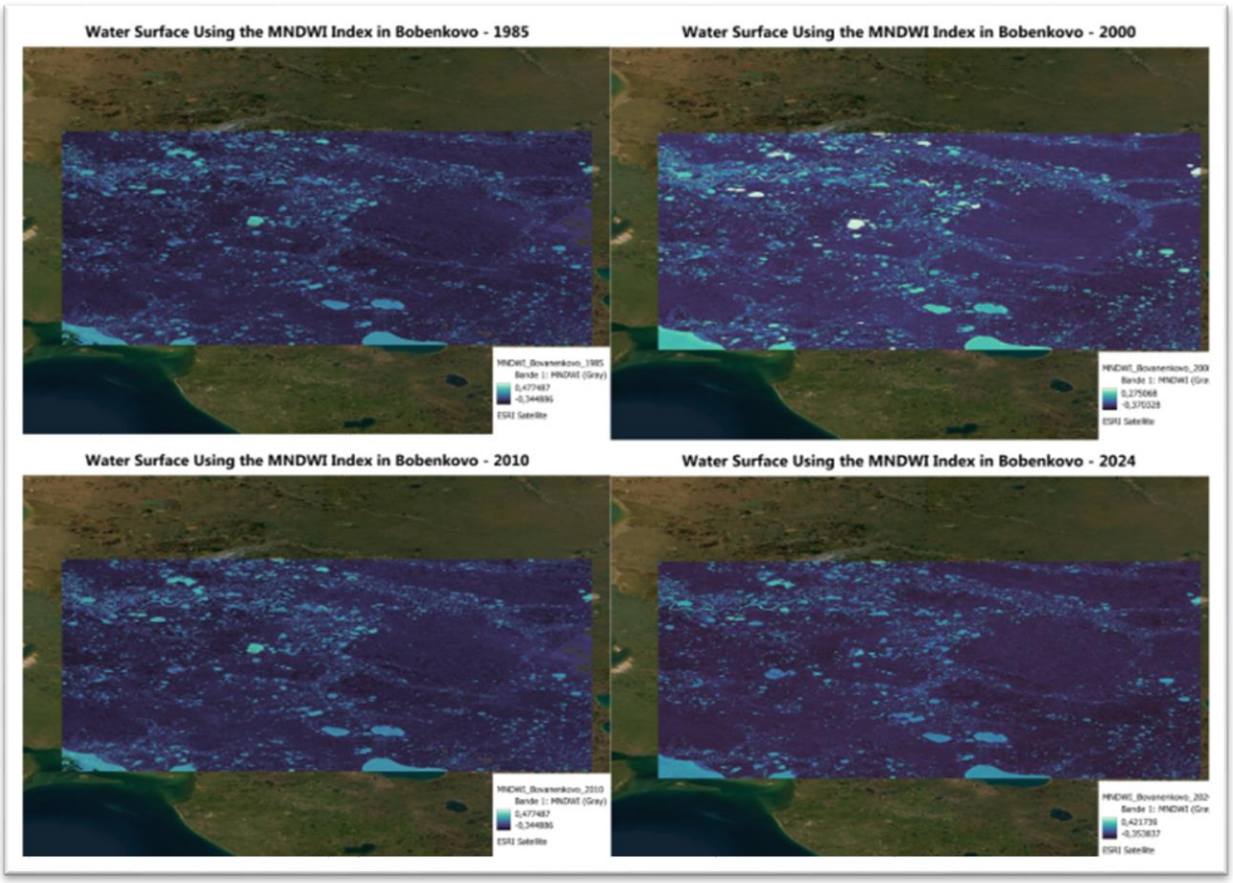


Figure 4-1. Multi-temporal evolution of water surfaces detected by MNDWI (Sentinel-2) across five key dates (1985–2024). Threshold: MNDWI > 0.20. Coordinate system: UTM Zone 42N.

Results:

A net increase in small ponds is observed in the lowlands, with lake expansion in areas with limited anthropogenic activity. Localized drainage is detected in the vicinity of roads and embankments, reflecting anthropogenic forcing on micro-hydrology. LST–MNDWI co-variability confirms the interpretation: pixels corresponding to water and very wet soils present lower heat signatures than mineral surfaces.

Interpretation:

The progression of water surfaces reflects active permafrost degradation through thermokarst dynamics. The formation of new thermokarst lakes in peripheral areas is driven by natural permafrost thaw acceleration, consistent with the NDWI trends documented in Section 3C. Conversely, the drainage of water bodies near industrial infrastructure testifies to anthropogenic forcing on local hydrology through embankments, drainage channels, and road construction that disrupt the natural surface water flow.

B) Vegetation Health (NDVI Analysis)

Methodology:

- Data: Sentinel-2 L2A (B8, B4) and Landsat (B4, B3) in summer
- SCL cloud mask; reprojection EPSG:32642; harmonization of spatial extent
- NDVI = (B8 − B4) / (B8 + B4)
- Classification: NDVI < 0.2 = stress / bare soil; 0.2–0.4 = tundra; > 0.4 = healthy vegetation

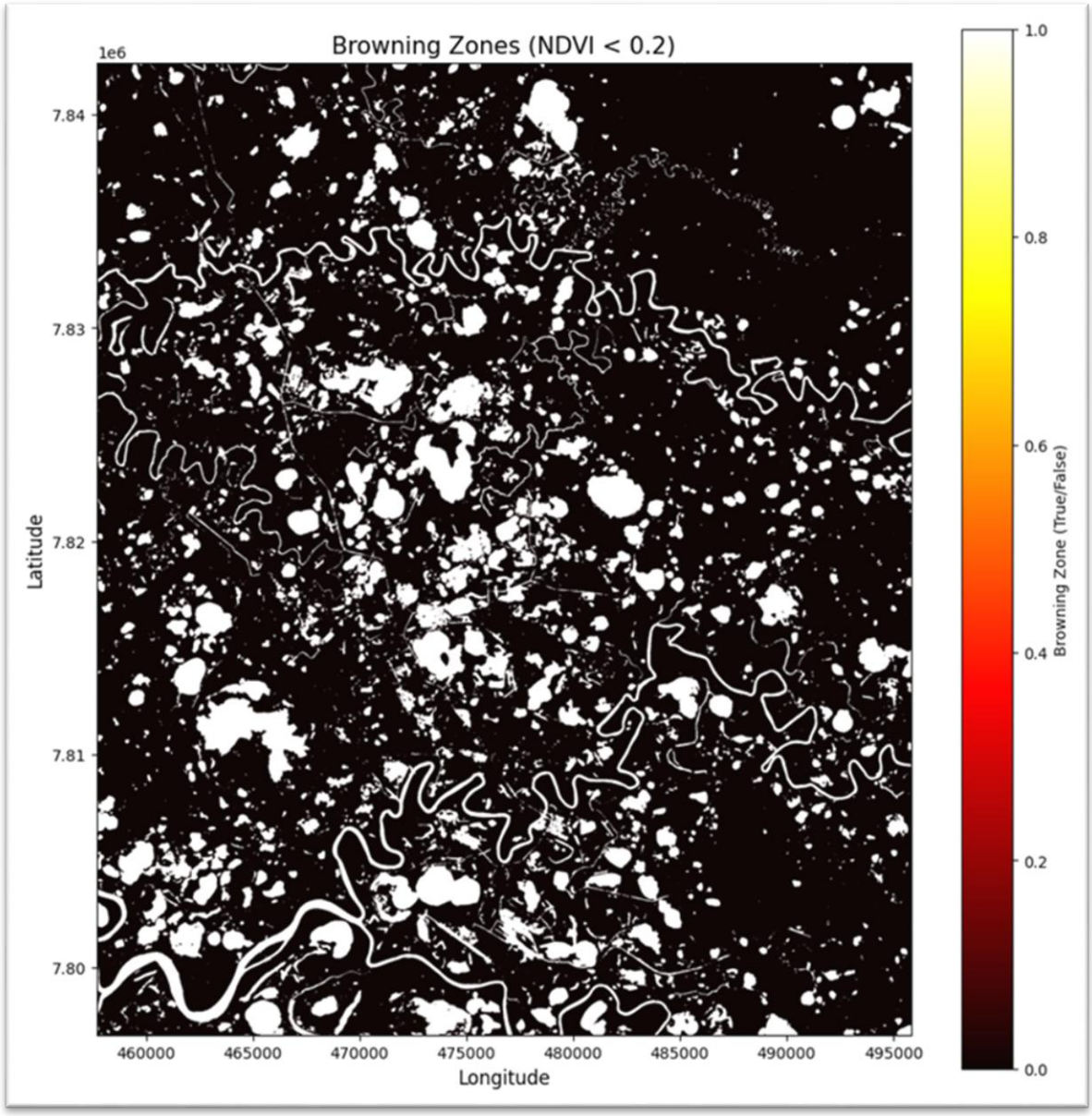


Figure 4-2. NDVI classified map of the Bovanenkovo area. Classes: stress/bare soil (NDVI < 0.2, red), tundra (0.2–0.4, yellow), healthy vegetation (> 0.4, green).

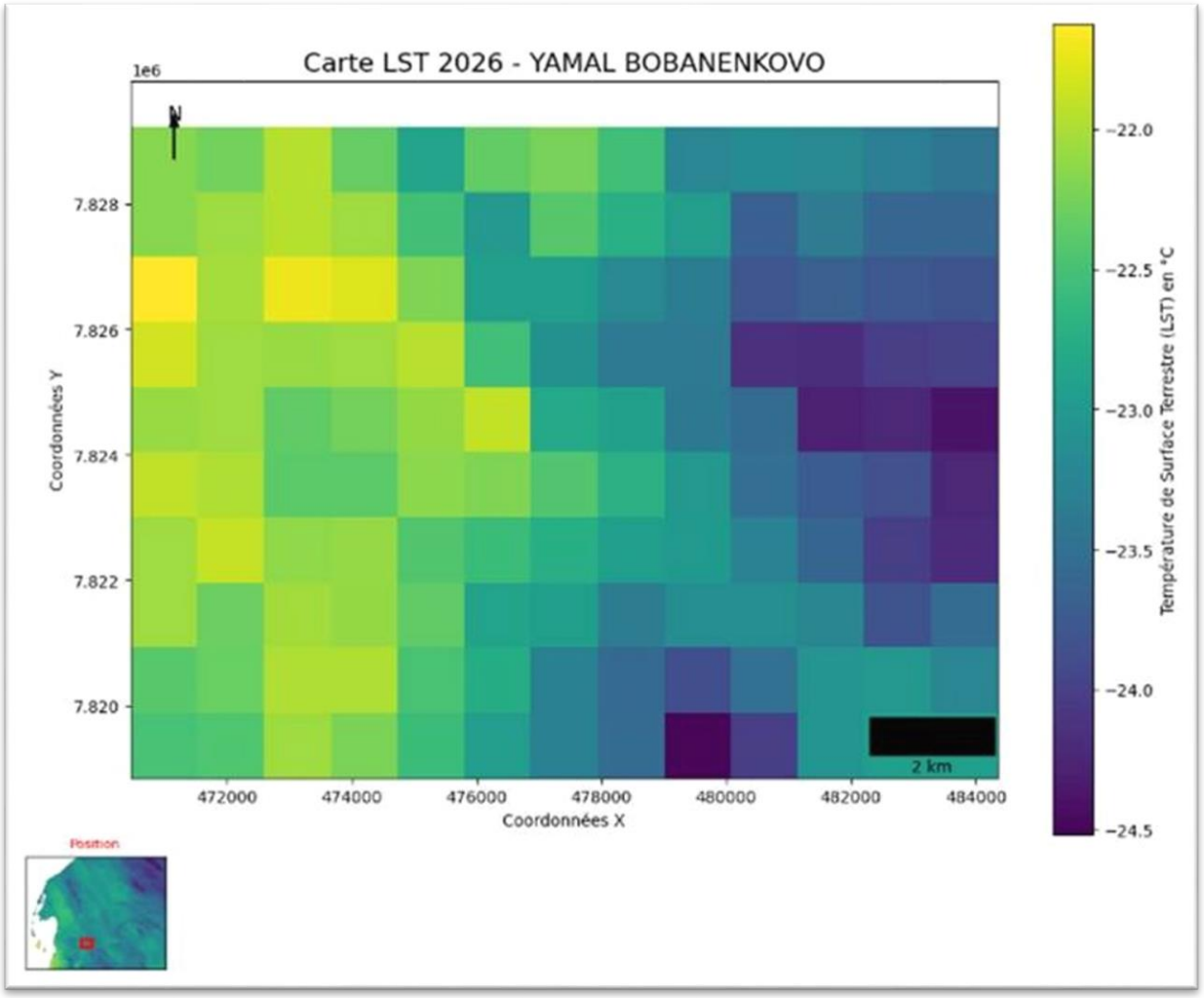


Figure 4-3. Land Surface Temperature map (Sentinel-3 SLSTR L2, 2026). Temperature in °C (conversion from Kelvin). Warmer areas (orange-red) correspond to mineral/dry surfaces; cooler areas (blue-violet) correspond to water/wet surfaces. UTM Zone 42N projection.

Results:

A browning ring of low NDVI values is visible around runways, gas extraction platforms, and the rail corridor. A pronounced negative NDVI deviation is observed within 250 m of infrastructure, narrowing progressively beyond 500 m. The mean NDVI shows an unstable long-term trend: strong growth between 1985 and 1990, followed by a decline into the early 2000s, a partial recovery, and then marked interannual variability since 2015, indicative of tundra vegetation that is highly sensitive to hydrological and thermal permafrost fluctuations.

Interpretation:

Vegetation stress is induced by dust deposition, soil compaction, and hydrological changes linked to infrastructure. Low NDVI zones frequently coincide with higher LST signatures and Sentinel-1 surface perturbation anomalies, confirming the convergence of thermal, SAR, and optical indicators of anthropogenic disturbance.

Note on Atmospheric Pollutant Analysis (Sentinel-5P TROPOMI):

The NO_2 concentration map derived from Sentinel-5P TROPOMI shows generally very low values throughout the Arctic zone, as expected for a low-emission region. However, a faint signal is detectable over Bovanenkovo, indicating slightly elevated background emissions consistent with the presence of the gas complex, where possible sources include turbine combustion, occasional flares, and local industrial traffic. The intensity remains low, but the point detection corresponds well to the known location of the gas field.

The CH_4 (methane) concentration map shows generally high values throughout the southern zone — typical of natural emissions from permafrost, wetlands, and Arctic peatlands. A slight intensification is observed north of the Yamal area, consistent with the gas corridor (infrastructure, extraction, and transport), although the intensity remains moderate. Unlike NO_2, CH_4 exhibits more diffuse and extensive spatial patterns due to atmospheric dispersion and dominant natural emissions.

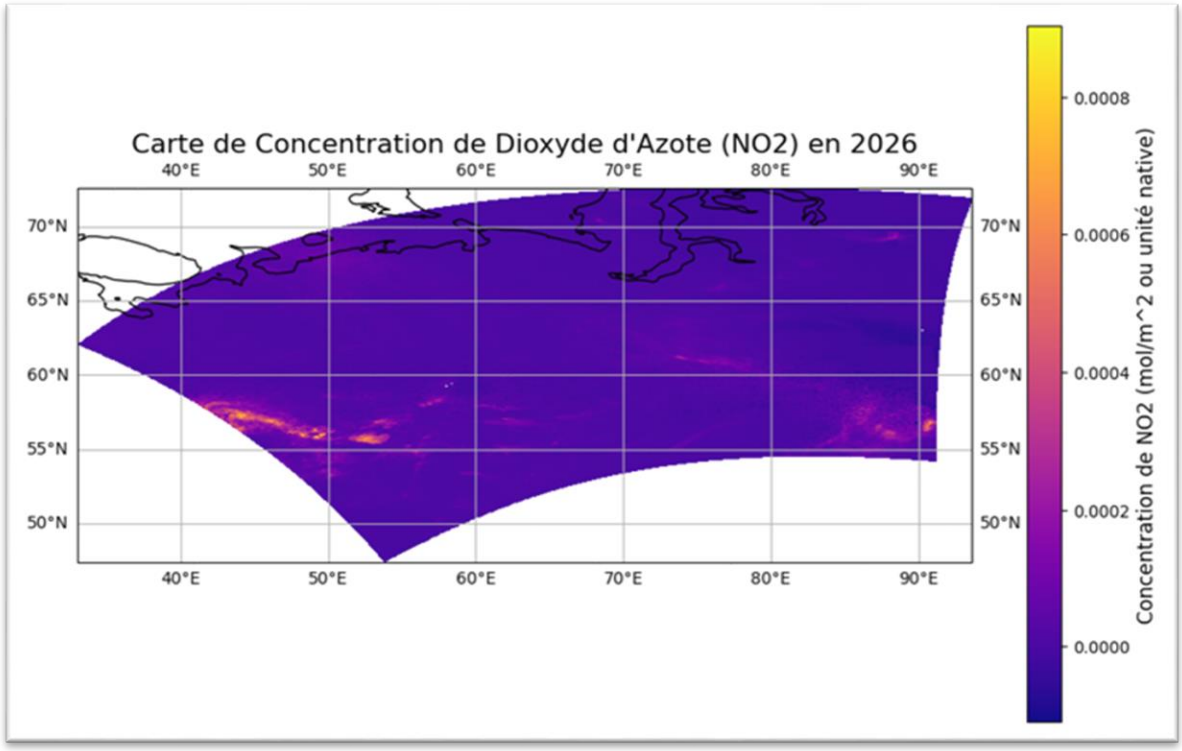


Figure 4-4. Tropospheric NO_2 column density (Sentinel-5P TROPOMI). The faint signal over Bovanenkovo (indicated) is consistent with industrial combustion activity.

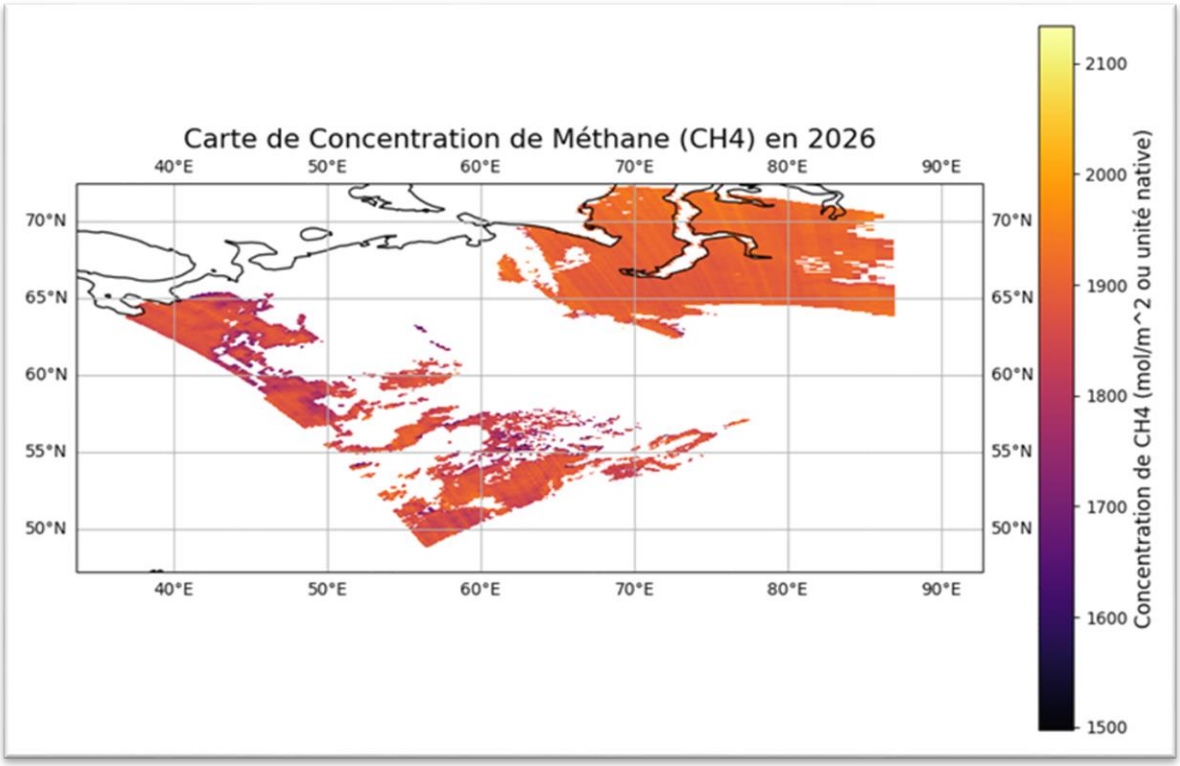


Figure 4-5. Methane (CH_4) column concentration (Sentinel-5P TROPOMI). The elevated values reflect both natural permafrost-related emissions and the diffuse signal of the gas infrastructure corridor.

5. Discussion

A) Correlation: Gas Production vs. Territorial Expansion

This section analyses the evolution of gas production at the Bovanenkovo site between 2011 and 2023, based on official Gazprom data, and compares it with the built-up surface area derived from Landsat 5 and 8 imagery classification.

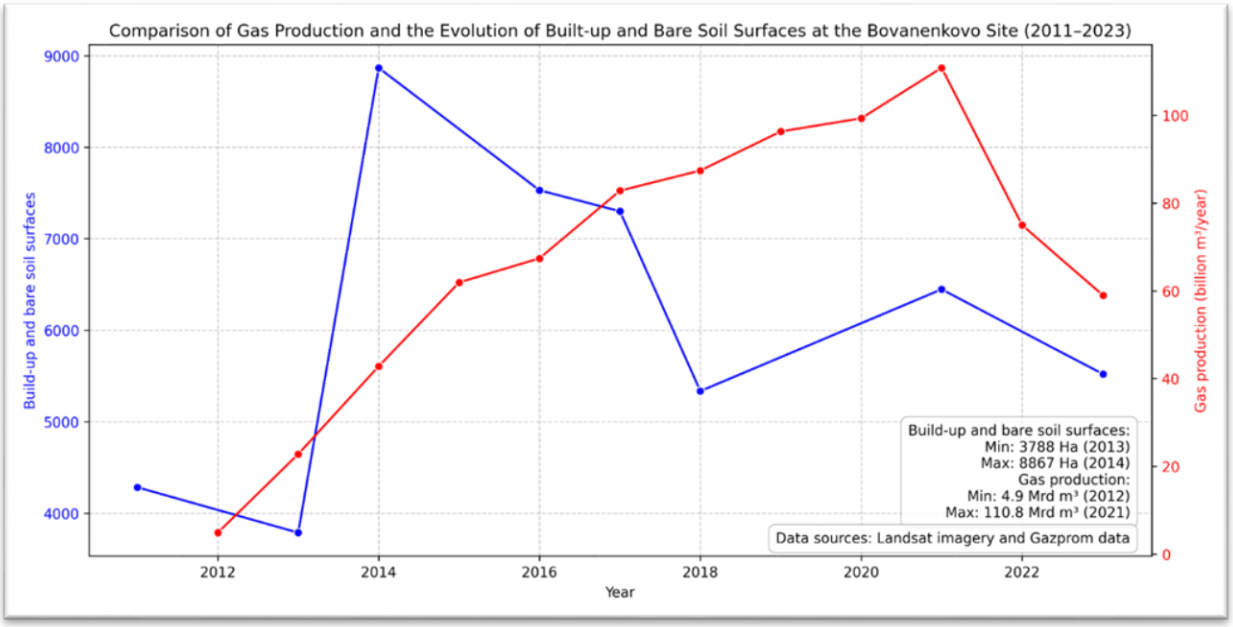


Figure 5-1. Comparison of annual gas production (Gazprom, bcm) and classified built-up surface area (Landsat, Ha) at Bovanenkovo between 2011 and 2023.

Regarding built-up surface evolution, it should be noted that the classification combines built-up areas and bare soils into a single class, as distinguishing between these two surface types proved extremely difficult given their spectral similarity. A significant increase in this combined class is observed between 2013 and 2014, which can be directly linked to the major northward expansion of gas extraction infrastructure documented in Section 3A. After 2014, the graph shows apparent fluctuations — including apparent decreases — which are attributable to the over- and under-representation of bare soils in the classification at different dates rather than a genuine reduction in infrastructure. This curve therefore serves as an indicator of overall trends rather than a precise measurement: an increase up to 2014, followed by stagnation, consistent with the more detailed cartographic analysis of Section 3A.

Regarding gas production, output began in 2012 and grew steadily until 2022, when a sharp decline occurred. This drop is primarily attributable to major geopolitical shifts: Russia's invasion of Ukraine in February 2022, followed by the European Union's strategy to reduce dependence on Russian gas from March–May 2022. The EU represented 64% of Russian gas exports in 2021, according to the International Energy Agency. The situation was compounded by the explosion of the Nord Stream pipeline in 2022 — which had transported 38% of Russian gas destined for Europe in 2021 — effectively ending that transit route.

In response to this contraction of exports to Europe, Russia has accelerated its gas export strategy towards Asia, primarily through the “Power of Siberia 2” pipeline project connecting Siberian gas fields to China, following its predecessor linking western Siberia to the Chinese market. This geopolitical pivot is expected to shape the future trajectory of Bovanenkovo’s production in the coming decade.

The correlation between territorial expansion and production volumes confirms a clear phased pattern: infrastructure buildup preceded production ramp-up, and the spatial stabilization of the industrial footprint after 2014 was followed by sustained and growing production volumes until the geopolitical rupture of 2022. This sequential relationship between territorial organization and economic output underlines the extent to which the transformation of the Arctic landscape was purposefully and systematically engineered to serve industrial objectives.

B) Limitations & Uncertainties

While the multi-source remote sensing methodology applied in this study provides valuable insights into the industrial and environmental transformation of the Bovanenkovo gas field, several technical and contextual limitations must be acknowledged in order to properly interpret the results.

Spatial Resolution Constraints:

The primary satellite used for infrastructure detection, Sentinel-1 SAR, operates at a ground resolution of approximately 10 to 40 metres depending on the acquisition mode. While this resolution is sufficient to identify large industrial structures such as drilling pads, processing facilities, and major transport axes, it proves inadequate for mapping finer landscape elements. Minor access roads, individual pipelines, and small surface disturbances cannot be reliably detected or quantified at this scale. Similarly, Sentinel-3 SLSTR, used for Land Surface Temperature mapping, offers a coarse resolution of approximately 1 kilometre, which limits its ability to capture localized thermal anomalies associated with specific infrastructure elements or permafrost degradation hotspots.

Optical Data Availability and Temporal Coverage:

Sentinel-2 multispectral imagery, offering 10-metre resolution well suited to vegetation and water index analysis, only became operational from 2015 onwards. For the earlier phases of the study period — particularly the critical 1988–2014 construction and pre-production decades — the analysis relied on Landsat imagery. This shift introduces inconsistencies in spatial resolution and spectral comparability across the full temporal record. The Arctic climate of the Yamal Peninsula further restricts the availability of cloud-free and snow-free acquisitions, as the study area remains snow-covered for approximately 240 days per year. This constraint was especially acute between 2000 and 2010, where usable Landsat scenes were particularly scarce. The failure of the Landsat 7 Scan Line Corrector in May 2003, which generated systematic diagonal data gaps, further degraded scene quality during this decade.

Classification Accuracy:

The unsupervised ISOCLUST classification method, while practical and adaptable, does not incorporate prior knowledge of target class spectral signatures. In the context of Bovanenkovo, where industrial surfaces and natural bare soils share very similar spectral responses in optical imagery, this approach has clear limitations. A combined

use of optical and SAR data would have helped reduce this spectral confusion. Furthermore, replacing ISOCLUST with a supervised classification model — such as Random Forest trained on labelled land cover samples — could have significantly improved precision and reliability, especially in ambiguous transition zones between industrial pads and surrounding tundra.

Atmospheric and Environmental Noise:

Cloud cover remains a recurrent issue with optical data, and despite the use of SCL masks, residual contamination may still affect NDVI and MNDWI calculations. Sentinel-1 SAR data, while theoretically immune to clouds, remains sensitive to variations in soil moisture, snow depth, and surface roughness, causing backscatter fluctuations that can be confused with anthropogenic signals across different seasons.

Ground Truth and Field Validation:

The extreme remoteness of the Yamal Peninsula made in-situ verification of classification results practically impossible. Without field data, formal accuracy assessment using standard metrics such as a confusion matrix or Kappa coefficient could not be conducted. Interpretations therefore rely on the internal consistency of multi-sensor results and their agreement with available cartographic and statistical references.

Despite these limitations, the convergence of results across multiple data sources — Sentinel-1, Sentinel-2, Sentinel-3, Landsat, and Sentinel-5P — strengthens the overall reliability of the conclusions drawn in this report, and highlights clear methodological pathways for improvement in future monitoring efforts of Arctic industrial territories.

6. Conclusion & Future Projections

A) General Conclusion

This study has demonstrated how the Bovanenkovo gas field has fundamentally transformed the Arctic landscape of the Yamal Peninsula over the period from 1985 to 2026. Through a multi-source remote sensing and GIS approach — combining Sentinel-1 SAR, Sentinel-2 multispectral imagery, Sentinel-3 LST, Landsat, ArcticDEM, and atmospheric data from Sentinel-5P — several key findings have emerged.

From a territorial perspective, built-up infrastructure was virtually absent in 1990 but expanded dramatically following the major construction phase initiated by Gazprom from 2008 onwards. By 2014, the industrial footprint had extended significantly northward, encompassing roads, drilling pads, residential facilities, logistics infrastructure, an airport, and a railway station. Thereafter, the rate of spatial expansion largely stabilized, although production volumes continued to increase until 2022. The Sentinel-1 radar analysis confirms a hierarchical and centralized industrial system, where the Obskaya–Bovanenkovo railway acts as the primary spatial backbone, organizing industrial expansion and connectivity across the territory.

From an environmental perspective, the multi-temporal analysis of spectral indices (NDWI, NDMI, NDVI) identifies a critical tipping point between 2011 and 2017, characterized by the rapid expansion of thermokarst water surfaces and a simultaneous decline in soil moisture indicators. The Combined RiskMAP confirms that permafrost degradation is most acute in the central and southern portions of the study area — precisely where infrastructure density is highest — highlighting a self-reinforcing feedback between industrial development and cryospheric instability.

The economic correlation analysis reveals that infrastructure buildup preceded production ramp-up, confirming the instrumental nature of the landscape transformation. The sharp production decline observed from 2022 reflects an exogenous geopolitical rupture rather than any natural or environmental threshold, underscoring the degree to which the Arctic transformation documented here is ultimately driven by global energy markets and political dynamics.

In response to the central research question, this study confirms that the industrial expansion of the Bovanenkovo gas field has profoundly and measurably transformed the Arctic landscape of the Yamal Peninsula through both the direct occupation of territory by infrastructure and the indirect amplification of environmental pressures on fragile permafrost and tundra ecosystems. The transformation is progressive, spatially organized, and increasingly irreversible at a human timescale.

B) Future Outlook via Scenario Analysis (2026–2050)

Climate projections suggest that Arctic regions will experience warming at a rate significantly higher than the global average. As a result, permafrost in the Yamal Peninsula is expected to undergo accelerated degradation during the coming decades.

Between 2026 and 2050, progressive ground warming could increase the thickness of the active layer and intensify thermokarst development. Ice-rich permafrost soils may

lose structural stability as ground ice melts, producing uneven surface subsidence and the formation of new thermokarst depressions. The industrial infrastructure of the Bovanenkovo gas field — including pipelines, railways, and drilling platforms — may face increasing structural risks, since much of this infrastructure relies on the mechanical stability of frozen ground.

Three possible scenarios can be considered:

Scenario 1 — Moderate Warming:

Thermokarst activity increases gradually, but infrastructure adaptation measures — such as elevated foundations, thermosyphon cooling systems, and improved drainage — allow continued operation with manageable risks and moderate additional costs.

Scenario 2 — Accelerated Degradation:

Rapid permafrost thaw results in widespread ground subsidence and thermokarst expansion. Infrastructure maintenance costs rise significantly due to repeated damage to pipelines, road surfaces, and rail embankments. Extraction operations are increasingly disrupted.

Scenario 3 — High-Impact Collapse:

Abrupt permafrost collapse and methane-related crater formation occur more frequently. Large sections of industrial infrastructure may require relocation or major redesign to remain operational. Environmental damage becomes extensive and partially irreversible.

Given the trends documented in this study — particularly the critical tipping point observed between 2011 and 2017 and the acceleration of LST anomalies — Scenario 2 appears increasingly likely for the 2026–2050 horizon without significant mitigation measures. The challenge for Gazprom and Russian authorities will be to balance continued exploitation with structural adaptation of existing infrastructure to a rapidly warming and destabilizing permafrost environment.

Data Sources

- Sentinel-1 SAR GRD — European Space Agency (ESA) / Copernicus Programme. Available via: Google Earth Engine, Copernicus Open Access Hub.
- Sentinel-2 MSI L2A — ESA / Copernicus Programme. Bands B2, B3, B4, B8, B11. Available via: Google Earth Engine, Copernicus Open Access Hub.
- Sentinel-3 SLSTR L2 LST — ESA / Copernicus Programme. Available via: Copernicus Open Access Hub.
- Sentinel-5P TROPOMI — ESA / Copernicus Programme. Products: L2__NO2___ and L2__CH4___. Available via: Google Earth Engine.
- Landsat 5 TM, Landsat 7 ETM+, Landsat 8 OLI — USGS / NASA. Available via: USGS EarthExplorer, Google Earth Engine.
- ArcticDEM — Polar Geospatial Center, University of Minnesota. Available via: https://www.pgc.umn.edu/data/arcticdem/
- OpenStreetMap — OpenStreetMap contributors. Extracted via QuickOSM plugin (QGIS) and Overpass API.
- Gazprom Annual Reports — Official gas production statistics (bcm). Available via: https://www.gazprom.com/investors/reports/

References

Chuvilin, E., Bukhanov, B., Davletshina, D., Grebenkin, S., & Istomin, V. (2018). Dissociation and self-preservation of gas hydrates in permafrost. Geosciences, 8(12), 431.

Clark Labs. (2020). TerrSet Geospatial Monitoring and Modeling System: Manual and Tutorial. Clark University, Worcester, MA, USA.

Drusch, M., Del Bello, U., Carlier, S., Colin, O., Fernandez, V., Gascon, F., … & Martimort, P. (2012). Sentinel-2: ESA's optical high-resolution mission for GMES operational services. Remote Sensing of Environment, 120, 25–36.

ESA. (2025). Sentinel-2 User Handbook. European Space Agency, Paris, France.

Forbes, B.C., Stammler, F., Kumpula, T., Meschtyb, N., Pajunen, A., & Kaarlejärvi, E. (2009). High resilience in the Yamal-Nenets reindeer herders to socio-economic and land-use changes. Proceedings of the National Academy of Sciences, 106(52), 22041–22048.

Gadal, S. (2012). Geographic space ontology, locus-object, and spatial data representation semantic theory. In Information Resources Management Association (Ed.), Geographic Information Systems: Concepts, Methodologies, Tools, and Applications (pp. 28–52). IGI Global, Hershey, PA, USA.

Gloaguen, T., Zaleckis, K., & Gadal, S. (2025). Modelling the road network as an expression of historical spatial system change in the Klaipeda region (Lithuania). SCITEPRESS: Setúbal, Portugal (accepted).

Kumpula, T., Forbes, B.C., Stammler, F., & Meschtyb, N. (2012). Dynamics of a coupled system: multi-resolution remote sensing of vegetation and reindeer in Arctic periglacial environments. Remote Sensing, 4(3), 579–600.

USGS. (2025). Landsat 8 Data Users Handbook. U.S. Geological Survey, Reston, VA, USA.

Vivone, G., Alparone, L., Chanussot, J., Dalla Mura, M., Garzelli, A., Licciardi, G.A., … & Wald, L. (2015). A critical comparison of pansharpening algorithms for remote sensing images. IEEE Transactions on Geoscience and Remote Sensing, 53(5), 2291–2305.

Walker, D.A., Raynolds, M.K., Daniëls, F.J.A., Einarsson, E., Elvebakk, A., Gould, W.A., … & Yurtsev, B.A. (2005). The circumpolar Arctic vegetation map. Journal of Vegetation Science, 16(3), 267–282.

Weng, Q. (2012). Remote sensing of impervious surfaces in the urban areas: requirements, methods, and trends. Remote Sensing of Environment, 117, 34–49.